RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF INFORMATION TECHNOLOGY

CS23621- Mobile Application Development Laboratory

ACADEMIC YEAR
2025 – 2026

REGUALTION 2023
EVEN Semester

Year : III **Semester : VI**

TABLE OF CONTENTS

S.NO.	DATE	EXPERIMENT TITLE	MARK	SIGN.
1.		**Kotlin Basics**		
2.		**Functions**		
3		**Classes and Objects**		
4		**Build First Android App**		
5		**Layouts**		
6		**App Navigation**		
7		**Activity and Fragment Life Cycle**		
8		**App Architecture (UI LAYER)**		
9		**App Architecture (Persistence)**		
10		**Advanced Recycler View Use cases**		
11		**Connect to the Internet**		
12		**Repository Pattern and WorkManager**		
13.		**App UI Design**		

INTRODUCTION

What is Kotlin?
Kotlin is a modern, trending programming language that was released in 2016 by JetBrains. It has become very popular since it is compatible with Java (one of the most popular programming languages out there), which means that Java code (and libraries) can be used in Kotlin programs.

Kotlin is used for:

- Mobile applications (specially Android apps)
- Web development
- Server side applications
- Data science

Why Use Kotlin?

- Kotlin is fully compatible with Java
- Kotlin works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc.)
- Kotlin is concise and safe
- Kotlin is easy to learn, especially if you already know Java
- Kotlin is free to use
- Big community/support

Kotlin IDE
The easiest way to get started with Kotlin, is to use an IDE.

An IDE (Integrated Development Environment) is used to edit and compile code.

In this chapter, we will use IntelliJ (developed by the same people that created Kotlin) which is free to download from https://www.jetbrains.com/idea/download/.

Android's Kotlin-first approach

At Google I/O 2019, we announced that Android development will be increasingly Kotlin-first, and we've stood by that commitment. Kotlin is an expressive and concise programming language that reduces common code errors and easily integrates into existing apps. If you're looking to build an Android app, we recommend starting with Kotlin to take advantage of its best-in-class features.
In an effort to support Android development using Kotlin, we co-founded the Kotlin Foundation and have ongoing investments in improving compiler performance and build speed. To learn more about Android's commitment to being Kotlin-first, see Android's commitment to Kotlin.

Why is Android development Kotlin-first?
We reviewed feedback that came directly from developers at conferences, our Customer Advisory Board (CAB), Google Developer Experts (GDE), and through our developer research. Many developers already enjoy using Kotlin, and the request for more Kotlin support was clear. Here's what developers appreciate about writing in Kotlin:

- **Expressive and concise:** You can do more with less. Express your ideas and reduce the amount of boilerplate code. 67% of professional developers who use Kotlin say Kotlin has increased their productivity.

- **Safer code:** Kotlin has many language features to help you avoid common programming mistakes such as null pointer exceptions. Android apps that contain Kotlin code are 20% less likely to crash.
- **Interoperable:** Call Java-based code from Kotlin, or call Kotlin from Java-based code. Kotlin is 100% interoperable with the Java programming language, so you can have as little or as much of Kotlin in your project as you want.
- **Structured Concurrency:** Kotlin coroutines make asynchronous code as easy to work with as blocking code. Coroutines dramatically simplify background task management for everything from network calls to accessing local data.

What does Kotlin-first mean?
When building new Android development tools and content, such as Jetpack libraries, samples, documentation, and training content, we will design them with Kotlin users in mind while continuing to provide support for using our APIs from the Java programming language.

Components
Components of an Android Application
Introduction to Activities in Android
Services in Android with Example
Content Providers in Android with Example
Broadcast Receiver in Android With Example

TextView
TextView in Kotlin
Dynamic TextView in Kotlin
AutoCompleteTextView in Kotlin
Dynamic AutoCompleteTextView in Kotlin
CheckedTextView in Kotlin
Dynamic CheckedTextView in Kotlin

ScrollView
ScrollView in Android
HorizontalScrollView in Kotlin
Dynamic ScrollView in Kotlin
DynamicHorizontal ScrollView in Kotlin
Other Views

Dynamic ImageView in Kotlin
Android ListView in Kotlin
CardView in Android With Example
VideoView in Kotlin
Zigzag View in Android
ViewAnimator in Android with Example

Button
Button in Kotlin
RadioButton in Kotlin
ImageButton in Kotlin
ToggleButton in Kotlin
RadioGroup in Kotlin
CheckBox in Kotlin
Dynamic ImageButton in Kotlin
Dynamic RadioButton in Kotlin

EditText
Android EditText in Kotlin
Dynamic EditText in Kotlin

Layouts
Layouts in Android UI Design
Android UI Layouts
Android TableLayout in Kotlin
Android FrameLayout in Kotlin
Android RelativeLayout in Kotlin
Android ConstraintLayout in Kotlin
Android LinearLayout in Kotlin
Android Motion Layout in Kotlin

Bar
SeekBar in Kotlin
Dynamic SeekBar in Kotlin
Discrete SeekBar in Kotlin
RatingBar in Kotlin
Dynamic RatingBar in Kotlin
ProgressBar in Kotlin
Dynamic ProgressBar in Kotlin

Switcher
Switch in Kotlin
Dynamic Switch in Kotlin
TextSwitcher in Kotlin
Dynamic TextSwitcher in Kotlin
ImageSwitcher in Kotlin
Dynamic ImageSwitcher in Kotlin
Time and Date Picker
TimePicker in Kotlin

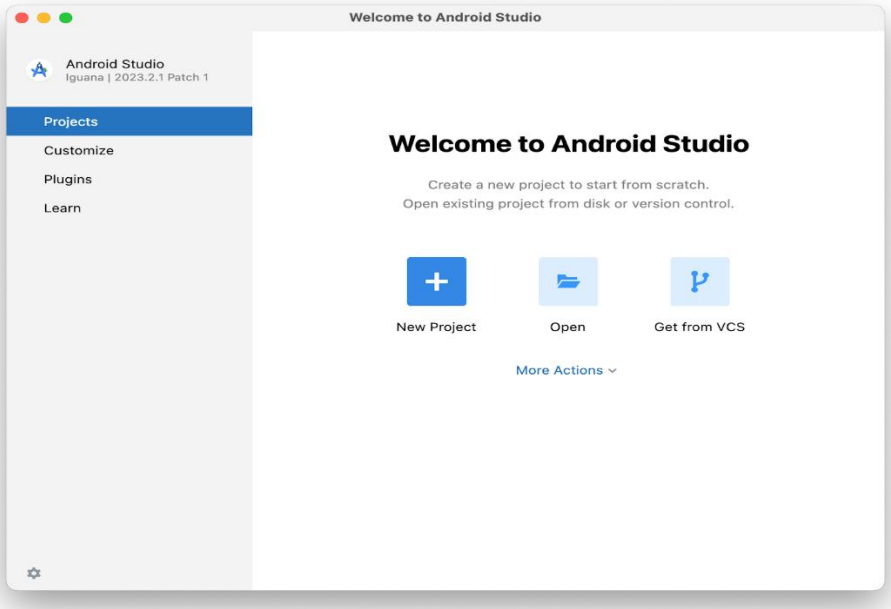
How to add a Custom-styled Toast in Android using Kotlin
Misc
Spinner
Dynamic Spinner in Kotlin
Advantage of Kotlin for Android Development
Choosing Kotlin for Android development offers several compelling benefits:

Concise Code: Kotlin lets you write less code to do the same things, making it easier to read and maintain.
Null Safety: Kotlin helps prevent common errors by making it harder to accidentally use null values, which can crash your app.
Interoperability with Java: You can use Kotlin and Java together in the same project. Kotlin works well with existing Java code, so you don't have to rewrite everything.
Modern Features: Kotlin has many modern programming features that make development faster and more enjoyable, like extension functions and data classes.
Better Performance: Kotlin often results in better-performing apps due to its efficient syntax and features.
Official Support: Kotlin is officially supported by Google for Android development, ensuring good documentation and ongoing improvements.
Active Community: A growing community of developers means lots of resources, libraries, and tools to help you learn and solve problems.

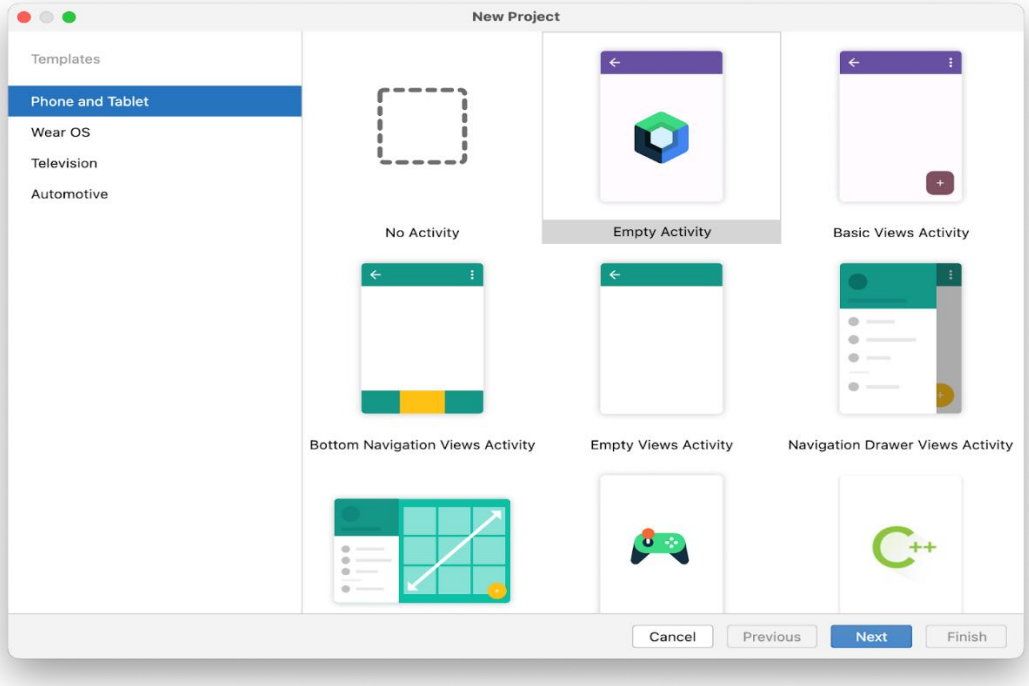
codelab, you create an Android app with the **Empty Activity** project template provided by Android Studio.
To create a project in Android Studio:

1. Double click the Android Studio icon to launch Android Studio.

2. In the **Welcome to Android Studio** dialog, click **New Project**.



The **New Project** window opens with a list of templates provided by Android Studio.

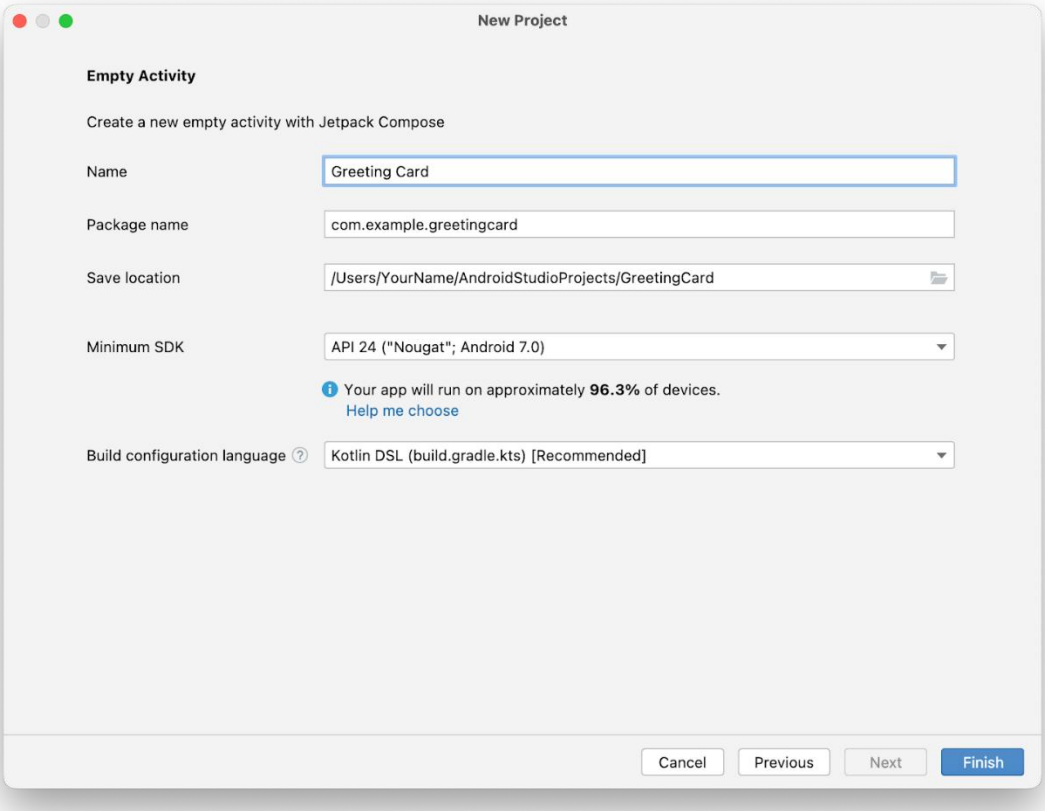


In Android Studio, a project template is an Android project that provides the blueprint for a certain type of app. Templates create the structure of the project and the files needed for Android Studio to build your project. The template that you choose provides starter code to get you going faster.

3. Make sure the **Phone and Tablet** tab is selected.
4. Click the **Empty Activity** template to select it as the template for your project. The **Empty Activity** template is the template to create a simple project that you can use to build a Compose app. It has a single screen and displays the text "Hello Android!".
5. Click **Next**. The **New Project** dialog opens. This has some fields to configure your project.
6. Configure your project as follows:

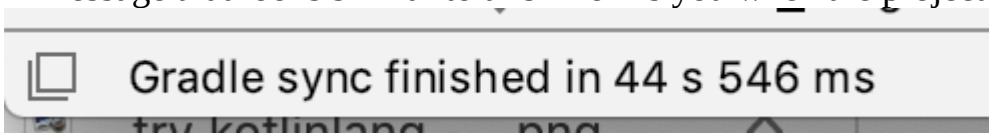
The **Name** field is used to enter the name of your project, for this codelab type "Greeting Card".

Leave the **Package name** field as is. This is how your files will be organized in the file structure. In this case, the package name will be com.example.greetingcard.
Leave the **Save location** field as is. It contains the location where all the files related to your project are saved. Take a note of where that is on your computer so that you can find your files.
Select **API 24: Android 7.0 (Nougat)** from the menu in the **Minimum SDK** field. Minimum SDK indicates the minimum version of Android that your app can run on.



7. Click **Finish**. This may take a while - this is a great time to get a cup of tea! While Android Studio is setting up, a progress bar and message indicates whether Android Studio is still setting up your project. It may look like this:

A message that looks similar to this informs you when the project set up is created.



8. You may see a **What's New** pane which contains updates on new features in Android Studio. Close it for now.

Assistant What's New

What's New in Iguana

This panel describes some of the new features and behavior changes included i
update.

To open this panel again later, select **Help > What's New in Android Studio** from
main menu.

Read in a browser

Version control system integration in App Quality Insights

Use **App Quality Insights** to navigate from a Crashlytics stack trace to the relev
code—at the point in time when the crash happened. When you view crash repo
choose to navigate to the line of code in your current git checkout or view a diff
between the current checkout and the version of your codebase that generated
crash.

Learn more

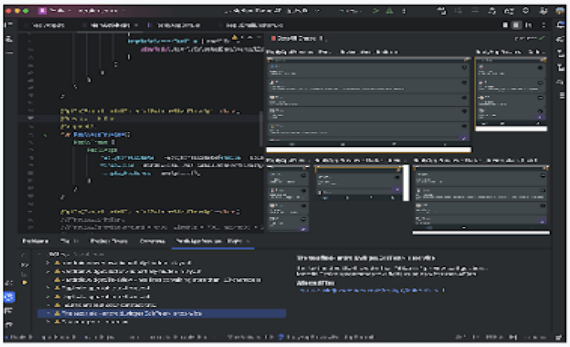
View Crashlytics crash variants in App Quality Insights

To help you analyze the root causes of a crash, use **App Quality Insights** to view
by issue *variants*, or groups of events that share similar stack traces.

Learn more

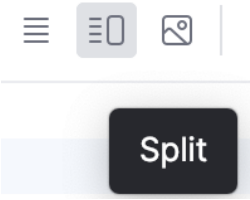
UI Check Mode in Compose Preview

To help developers build more adaptive and accessible UI in Jetpack Compose,
Studio Iguana Canary 5 introduced a new UI Check mode in Compose Preview.
feature works similar to Visual Linting and Accessibility checks integrations for
Activate Compose UI check mode for Android Studio to automatically audit your
Compose UI and check for adaptive and accessibility issues across different sc
sizes, such as text stretched on large screens and low color contrast. The mode
highlights issues found in different Preview configurations and lists them in the
problems panel.

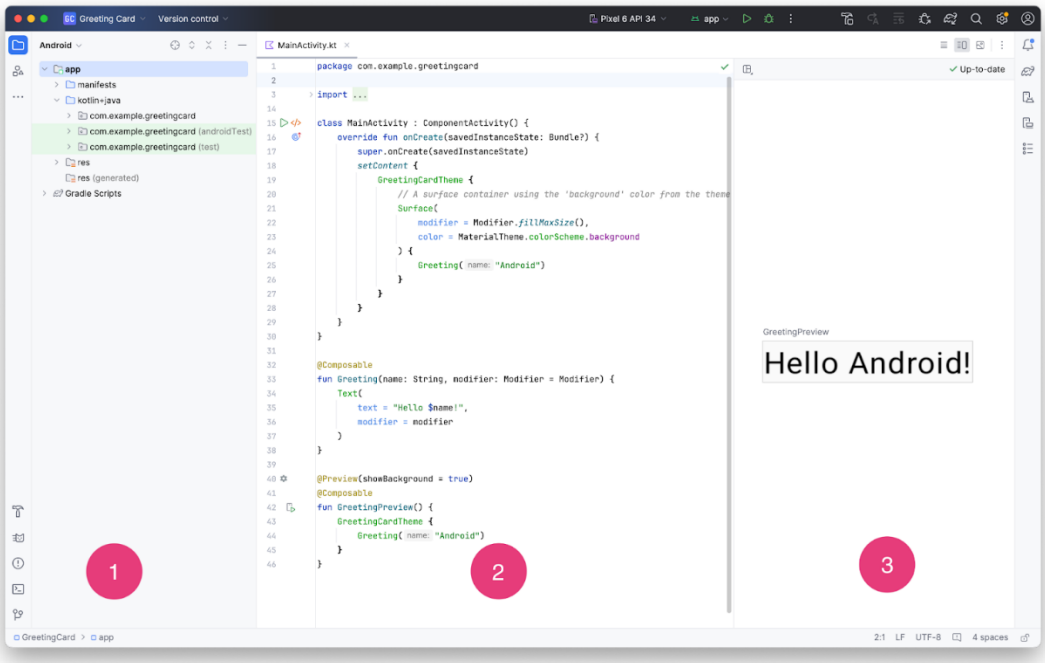


9. Click **Split** on the top right of Android Studio, this allows you to view both code and design. You can also click **Code** to view code only or click **Design** to view

design only.



After pressing **Split** you should see three areas:



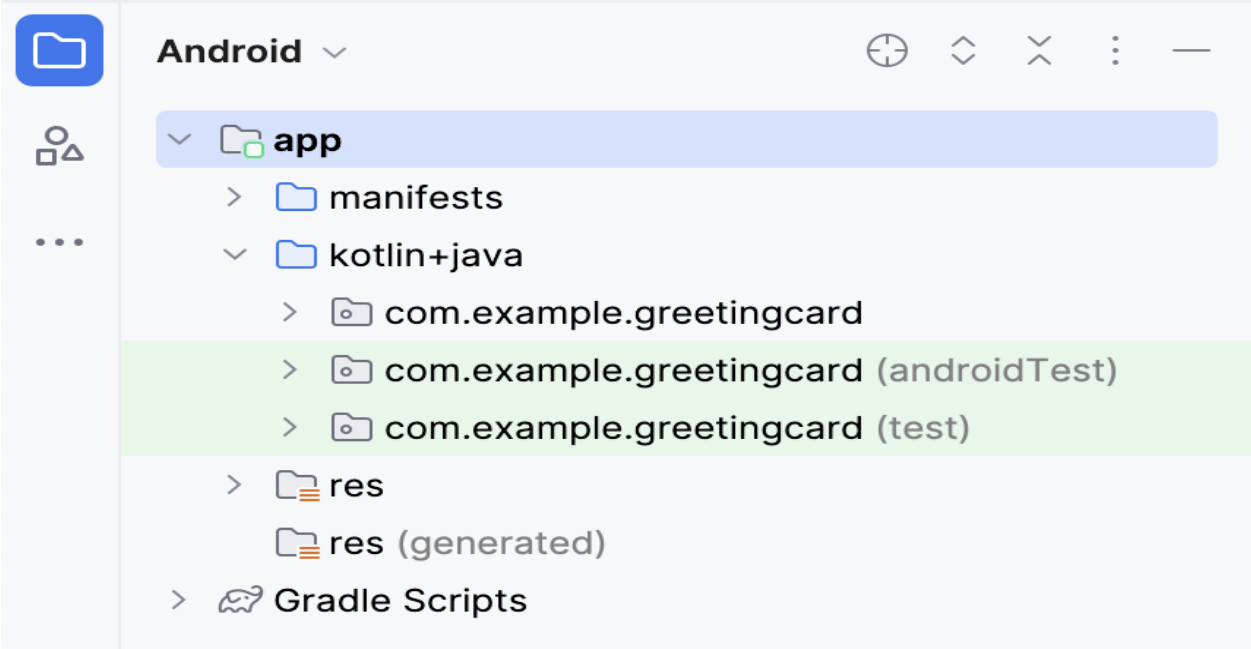
- The **Project** view (1) shows the files and folders of your project
- The **Code** view (2) is where you edit code
- The **Design** view (3) is where you preview what your app looks like

In the **Design** view, you may see a blank pane with this text:

10. Click **Build & Refresh**. It may take a while to build but when it is done the preview shows a text box that says "**Hello Android!**". Empty Compose activity contains all the code necessary to create this app.

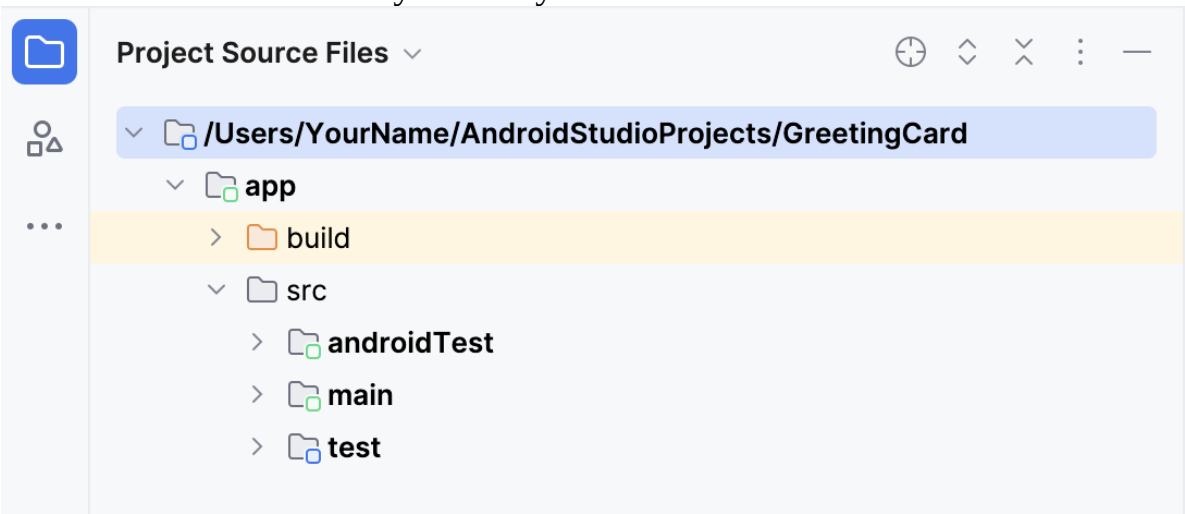
explore Android Studio by becoming familiar with the file structure.

1. In Android Studio, take a look at the **Project** tab. The **Project** tab shows the files and folders of your project. When you were setting up your project the package name was **com.example.greetingcard**. You can see that package right here in the **Project** tab. A package is basically a folder where code is located. Android Studio organizes the project in a directory structure made up of a set of packages.
2. If necessary, select **Android** from the drop-down menu in the **Project** tab.



This is the standard view and organization of files that you use. It's useful when you write code for your project because you can easily access the files you will be working on in your app. However, if you look at the files in a file browser, such as Finder or Windows Explorer, the file hierarchy is organized very differently.

3. Select **Project Source Files** from the drop-down menu. You can now browse the files in the same way as in any file browser.



4. Select **Android** again to switch back to the previous view. You use the **Android** view for this course. If your file structure ever looks strange, check to make sure you're still in **Android** view.

Code view of the MainActivity.kt file. Notice there are some automatically generated functions in this code, specifically the onCreate() and the setContent() functions.
Note: Remember that a function is a segment of a program that performs a specific task.

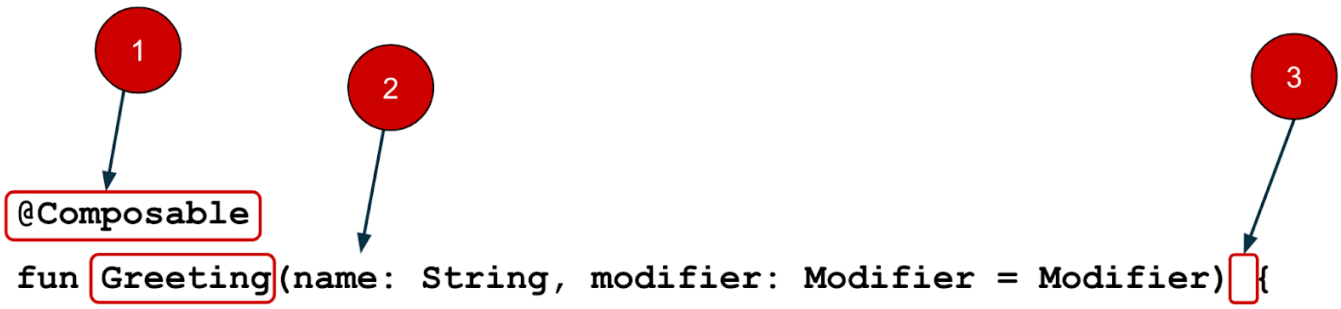
```
class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
        setContent {
            GreetingCardTheme {
                // A surface container using the 'background' color from the theme
                Surface(
                    modifier = Modifier.fillMaxSize(),
                    color = MaterialTheme.colorScheme.background
                ) {
                    Greeting("Android")
                }
            }
        }
    }
}
```

The onCreate() function is the entry point to this Android app and calls other functions to build the user interface. In Kotlin programs, the main() function is the entry point/starting point of execution. In Android apps, the onCreate() function fills that role.
The setContent() function within the onCreate() function is used to define your layout through composable functions. All functions marked with the @Composable annotation can be called from the setContent() function or from other Composable functions. The annotation tells the Kotlin compiler that this function is used by Jetpack Compose to generate the UI.
Note: The compiler takes the Kotlin code you wrote, looks at it line by line, and translates it into something that the computer can understand. This process is called compiling your code.
Next, look at the Greeting() function. The Greeting() function is a Composable function, notice the @Composable annotation above it. This Composable function takes some input and generates what's shown on the screen.

```
@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
}
```

You've learned about functions before (if you need a refresher, refer to the Create and use functions in Kotlin codelab), but there are a few differences with composable functions.



- 1. You add the @Composable annotation before the function

- 2. @Composable function names are capitalized.
- 3. @Composable functions can't return anything.

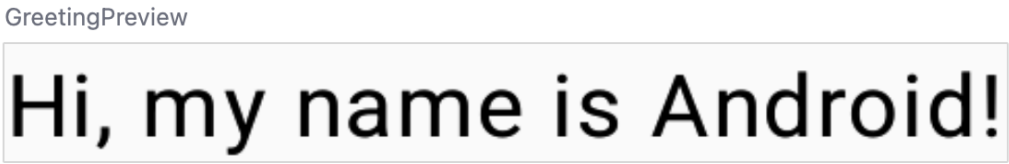
```
@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
}
```

Right now the Greeting() function takes in a name and displays Hello to that person.

1. Update the Greeting() function to introduce yourself instead of saying "Hello":

```
@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hi, my name is $name!",
        modifier = modifier
    )
}
```

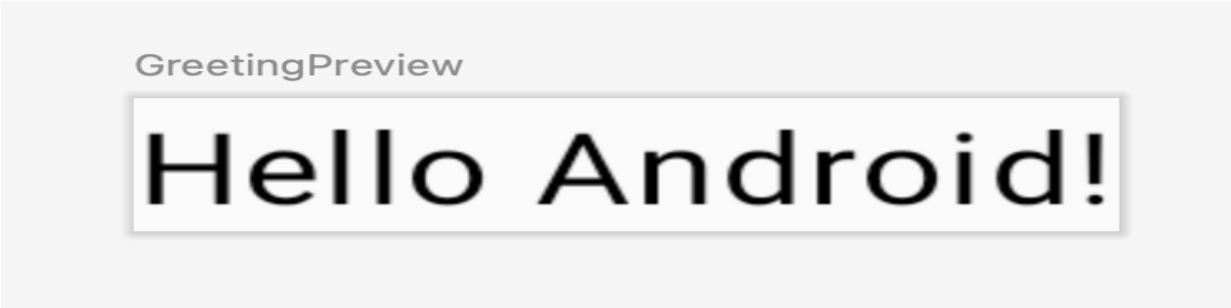
2. Android should automatically update the preview.



Great! You changed the text, but it introduces you as Android, which is probably not your name. Next, you will personalize it to introduce you with your name!

The GreetingPreview() function is a cool feature that lets you see what your composable looks like without having to build your entire app. To enable a preview of a composable, annotate with @Composable and @Preview. The @Preview annotation tells Android Studio that this composable should be shown in the design view of this file.

As you can see, the @Preview annotation takes in a parameter called showBackground. If showBackground is set to **true**, it will add a background to your composable preview. Since Android Studio by default uses a light theme for the editor, it can be hard to see the difference between showBackground = true and showBackground = false. However, this is an example of what the difference looks like. Notice the white background on the image set to true.



Expt No : 1 **Date:**

Kotlin Basics

Aim

To study and implement basic Kotlin programming concepts such as variables, data types, operators, control statements, and null safety.

Algorithm

1. Start the Android Studio IDE.
2. Create a new Kotlin project.
3. Declare variables using val and var.
4. Use basic data types such as Int, Double, String, and Boolean.
5. Implement conditional statements (if, when).
6. Implement loops (for, while).
7. Display output using println() or UI components.
8. Run the program and verify output.

Coding

```
package com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview
import com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b
fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!"
fun sumAll(vararg numbers: Int): Int = numbers.sum()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else factorial(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) {
```

```
    fun introduce(): String = "Hi, I'm $name and I'm $age years old"
}

data class User(val id: Int, val username: String, val email: String)

enum class Priority(val value: Int) {
    LOW(1), MEDIUM(2), HIGH(3), URGENT(4);
    fun getDisplayName(): String = name.lowercase().replaceFirstChar { it.uppercase() }
}

object AppConfig {
    const val APP_NAME = "Kotlin Demo"
    const val VERSION = "1.0.0"
    fun getConfig(): String = "$APP_NAME v$VERSION"
}

class MathUtils {
    companion object {
        const val PI = 3.14159
        fun circleArea(radius: Double): Double = PI * radius * radius
    }
}

abstract class Animal(val name: String) {
    abstract fun makeSound(): String
    fun sleep(): String = "$name is sleeping"
}

class Dog(name: String) : Animal(name) {
    override fun makeSound(): String = "$name says: Woof!"
}

interface Drawable {
    fun draw(): String
}

class Circle(private val radius: Double) : Drawable {
    override fun draw(): String = "Drawing a circle with radius $radius"
}

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Basic1Theme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    KotlinFundamentalsDemo(
```

```
                    modifier = Modifier.padding(innerPadding)
                )
            }
        }
    }
}

@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column(
        modifier = modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text = "Kotlin Fundamentals Demo",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") {
            demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
```

```
        }

        // Classes and Objects
        SectionCard(title = "Classes & Objects") {
            demonstrateClassesAndObjects()
        }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult ->
                result = newResult
            }
        }

        if (result.isNotEmpty()) {
            Card(
                modifier = Modifier.fillMaxWidth(),
                colors = CardDefaults.cardColors(
                    containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text(
                    text = "Result: $result",
                    modifier = Modifier.padding(16.dp),
                    fontSize = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(
        modifier = Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(
            modifier = Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text(
                text = title,
                fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
```

```
        }
    }
}

@Composable
fun demonstrateVariablesAndDataTypes() {
    // Immutable variable (val)
    val immutableName: String = "Kotlin"

    // Mutable variable (var)
    var mutableAge: Int = 10
    mutableAge += 1

    // Type inference
    val inferredDouble = 3.14159

    // Different data types
    val byteValue: Byte = 127
    val shortValue: Short = 32767
    val intValue: Int = 2147483647
    val longValue: Long = 9223372036854775807L
    val floatValue: Float = 3.14f
    val doubleValue: Double = 2.71828
    val charValue: Char = 'K'
    val booleanValue: Boolean = true
    val stringValue: String = "Hello Kotlin"
    val arrayValue: IntArray = intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val (immutable): $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type inference: $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long: $longValue")
        Text("Float: $floatValue")
        Text("Double: $doubleValue")
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}")
    }
}

@Composable
fun demonstrateOperators() {
    val a = 10
    val b = 3
```

```
    // Arithmetic operators
    val sum = a + b
    val difference = a - b
    val product = a * b
    val quotient = a / b
    val remainder = a % b

    // Comparison operators
    val isEqual = a == b
    val isNotEqual = a != b
    val isGreater = a > b
    val isLess = a < b

    // Logical operators
    val x = true
    val y = false
    val andResult = x && y
    val orResult = x || y
    val notResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text("  a + b = $sum")
        Text("  a - b = $difference")
        Text("  a * b = $product")
        Text("  a / b = $quotient")
        Text("  a % b = $remainder")

        Text("Comparison Operators:")
        Text("  a == b = $isEqual")
        Text("  a != b = $isNotEqual")
        Text("  a > b = $isGreater")
        Text("  a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text("  x && y = $andResult")
        Text("  x || y = $orResult")
        Text("  !x = $notResult")
    }
}

@Composable
fun demonstrateControlStatements() {
    val score = 85
    val grade = when {
        score >= 90 -> "A"
        score >= 80 -> "B"
```

```
        score >= 70 -> "C"
        score >= 60 -> "D"
        else -> "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5)
    val evenNumbers = mutableListOf<Int>()

    // For loop
    for (number in numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation
    var countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ")
            countdown--
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text("  Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text("  Numbers: ${numbers.joinToString()}")
        Text("  Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text("  Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    val nullableString: String? = "Hello"
    val nullString: String? = null

    // Safe call operator
    val length1 = nullableString?.length

    // Elvis operator
```

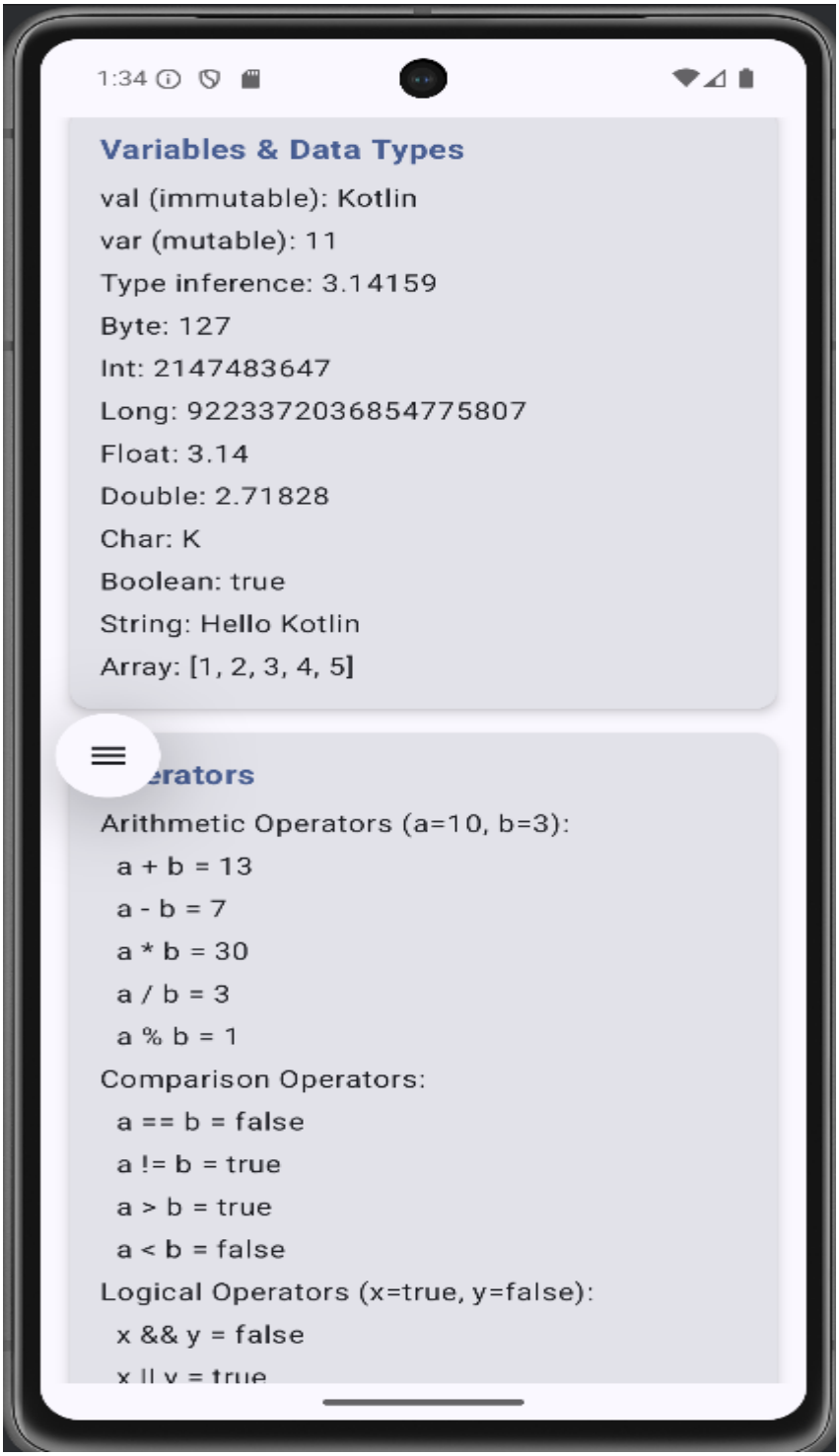
```
    val length2 = nullString?.length ?: 0

    // Non-null asserted call (use with caution)
    val length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin"
    val safeCast = anyValue as? String

    // Let scope function
    val result = nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() {
    // Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function
    fun String.isPalindrome(): Boolean {
        val cleaned = this.lowercase().filter { it.isLetter() }
        return cleaned == cleaned.reversed()
    }

    val sum = add(5, 3)
    val greeting = greet("Kotlin")
    val sumVarargs = sumAll(1, 2, 3, 4, 5)
    val product = operate(4, 5, multiply)
    val palindrome = "Racecar".isPalindrome()
    val fact = factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Basic Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product")
```

```
        Text("Extension: \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact")
    }
}

@Composable
fun demonstrateClassesAndObjects() {
    // Usage examples
    val person = Person("Alice", 25)
    val user = User(1, "john_doe", "john@example.com")
    val priority = Priority.HIGH
    val config = AppConfig.getConfig()
    val area = MathUtils.circleArea(5.0)
    val dog = Dog("Buddy")
    val circle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}")
    }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) {
    var input1 by remember { mutableStateOf("") }
    var input2 by remember { mutableStateOf("") }
    var operation by remember { mutableStateOf("add") }

    Column(verticalArrangement = Arrangement.spacedBy(8.dp)) {
        Text("Try it yourself!")

        OutlinedTextField(
            value = input1,
            onValueChange = { input1 = it },
            label = { Text("Number 1") },
            modifier = Modifier.fillMaxWidth()
        )

        OutlinedTextField(
            value = input2,
            onValueChange = { input2 = it },
            label = { Text("Number 2") },
            modifier = Modifier.fillMaxWidth()
```

```
        )

        Row(
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            modifier = Modifier.fillMaxWidth()
        ) {
            RadioButton(
                selected = operation == "add",
                onClick = { operation = "add" }
            )
            Text("Add")

            RadioButton(
                selected = operation == "subtract",
                onClick = { operation = "subtract" }
            )
            Text("Subtract")

            RadioButton(
                selected = operation == "multiply",
                onClick = { operation = "multiply" }
            )
            Text("Multiply")
        }

        Button(
            onClick = {
                val num1 = input1.toIntOrNull() ?: 0
                val num2 = input2.toIntOrNull() ?: 0
                val result = when (operation) {
                    "add" -> num1 + num2
                    "subtract" -> num1 - num2
                    "multiply" -> num1 * num2
                    else -> 0
                }
                onResult("$num1 $operation $num2 = $result")
            },
            modifier = Modifier.align(Alignment.CenterHorizontally)
        ) {
            Text("Calculate")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
    Basic1Theme {
```

```
        KotlinFundamentalsDemo()
    }
}
```

Output



Expt No : 2 **Date:**

Functions

Aim

To understand and implement different types of functions in Kotlin.

Algorithm

1. Create a Kotlin file.
2. Define a function using fun keyword.
3. Pass parameters to the function.
4. Specify return type.
5. Call the function from main() or activity.
6. Implement default arguments.
7. Execute the program.

Coding

```
package com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview
import com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b
fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!"
fun sumAll(vararg numbers: Int): Int = numbers.sum()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else factorial(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) {
```

```
    fun introduce(): String = "Hi, I'm $name and I'm $age years old"
}

data class User(val id: Int, val username: String, val email: String)

enum class Priority(val value: Int) {
    LOW(1), MEDIUM(2), HIGH(3), URGENT(4);
    fun getDisplayName(): String = name.lowercase().replaceFirstChar { it.uppercase() }
}

object AppConfig {
    const val APP_NAME = "Kotlin Demo"
    const val VERSION = "1.0.0"
    fun getConfig(): String = "$APP_NAME v$VERSION"
}

class MathUtils {
    companion object {
        const val PI = 3.14159
        fun circleArea(radius: Double): Double = PI * radius * radius
    }
}

abstract class Animal(val name: String) {
    abstract fun makeSound(): String
    fun sleep(): String = "$name is sleeping"
}

class Dog(name: String) : Animal(name) {
    override fun makeSound(): String = "$name says: Woof!"
}

interface Drawable {
    fun draw(): String
}

class Circle(private val radius: Double) : Drawable {
    override fun draw(): String = "Drawing a circle with radius $radius"
}

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Basic1Theme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    KotlinFundamentalsDemo(
```

```
                    modifier = Modifier.padding(innerPadding)
                )
            }
        }
    }
}

@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column(
        modifier = modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text = "Kotlin Fundamentals Demo",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") {
            demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
```

```
        }

        // Classes and Objects
        SectionCard(title = "Classes & Objects") {
            demonstrateClassesAndObjects()
        }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult ->
                result = newResult
            }
        }

        if (result.isNotEmpty()) {
            Card(
                modifier = Modifier.fillMaxWidth(),
                colors = CardDefaults.cardColors(
                    containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text(
                    text = "Result: $result",
                    modifier = Modifier.padding(16.dp),
                    fontSize = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(
        modifier = Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(
            modifier = Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text(
                text = title,
                fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
```

```
        }
    }
}

@Composable
fun demonstrateVariablesAndDataTypes() {
    // Immutable variable (val)
    val immutableName: String = "Kotlin"

    // Mutable variable (var)
    var mutableAge: Int = 10
    mutableAge += 1

    // Type inference
    val inferredDouble = 3.14159

    // Different data types
    val byteValue: Byte = 127
    val shortValue: Short = 32767
    val intValue: Int = 2147483647
    val longValue: Long = 9223372036854775807L
    val floatValue: Float = 3.14f
    val doubleValue: Double = 2.71828
    val charValue: Char = 'K'
    val booleanValue: Boolean = true
    val stringValue: String = "Hello Kotlin"
    val arrayValue: IntArray = intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val (immutable): $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type inference: $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long: $longValue")
        Text("Float: $floatValue")
        Text("Double: $doubleValue")
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}")
    }
}

@Composable
fun demonstrateOperators() {
    val a = 10
    val b = 3
```

```
    // Arithmetic operators
    val sum = a + b
    val difference = a - b
    val product = a * b
    val quotient = a / b
    val remainder = a % b

    // Comparison operators
    val isEqual = a == b
    val isNotEqual = a != b
    val isGreater = a > b
    val isLess = a < b

    // Logical operators
    val x = true
    val y = false
    val andResult = x && y
    val orResult = x || y
    val notResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text("  a + b = $sum")
        Text("  a - b = $difference")
        Text("  a * b = $product")
        Text("  a / b = $quotient")
        Text("  a % b = $remainder")

        Text("Comparison Operators:")
        Text("  a == b = $isEqual")
        Text("  a != b = $isNotEqual")
        Text("  a > b = $isGreater")
        Text("  a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text("  x && y = $andResult")
        Text("  x || y = $orResult")
        Text("  !x = $notResult")
    }
}

@Composable
fun demonstrateControlStatements() {
    val score = 85
    val grade = when {
        score >= 90 -> "A"
        score >= 80 -> "B"
```

```
        score >= 70 -> "C"
        score >= 60 -> "D"
        else -> "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5)
    val evenNumbers = mutableListOf<Int>()

    // For loop
    for (number in numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation
    var countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ")
            countdown--
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text("  Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text("  Numbers: ${numbers.joinToString()}")
        Text("  Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text("  Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    val nullableString: String? = "Hello"
    val nullString: String? = null

    // Safe call operator
    val length1 = nullableString?.length

    // Elvis operator
```

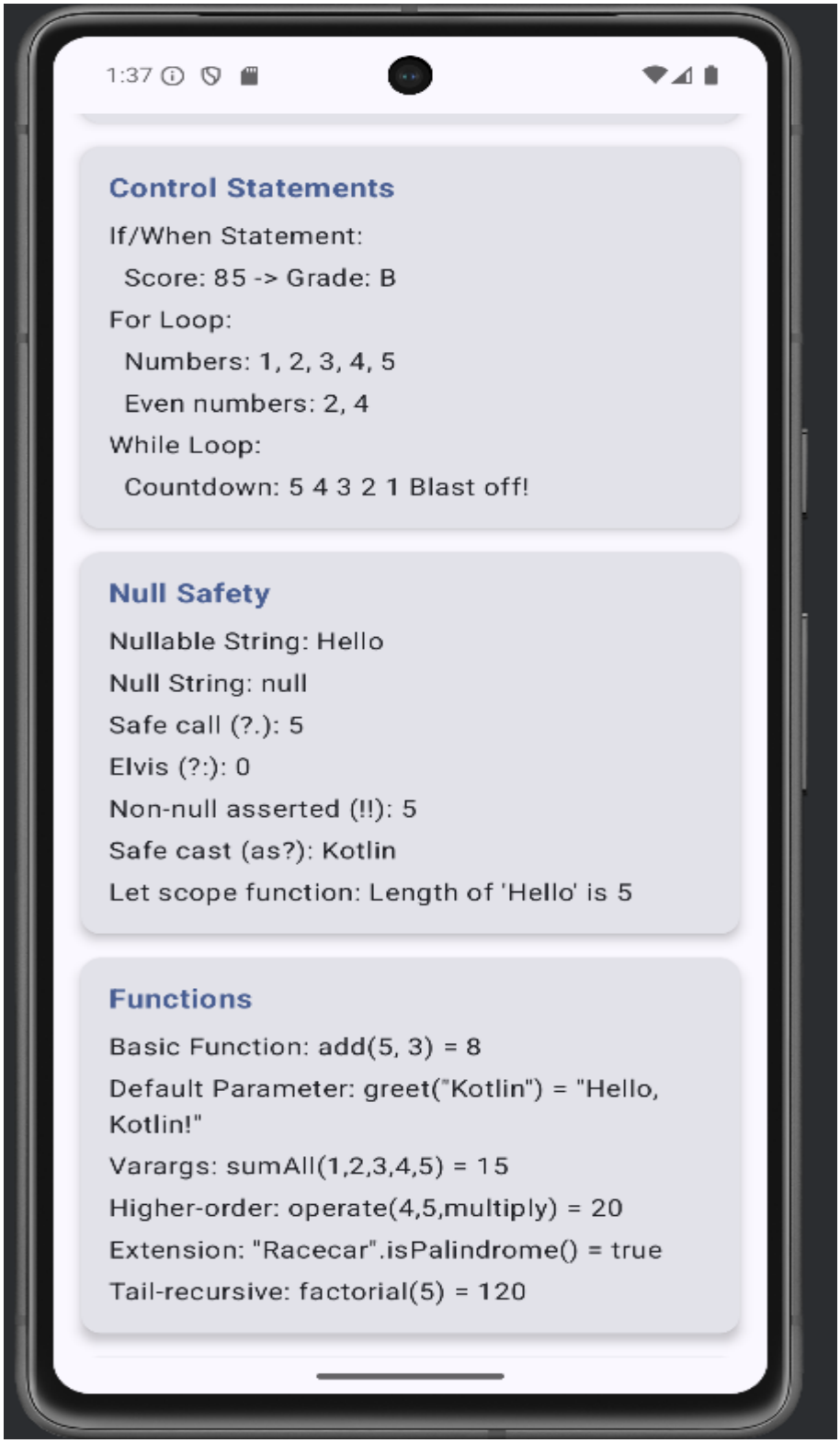
```
    val length2 = nullString?.length ?: 0

    // Non-null asserted call (use with caution)
    val length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin"
    val safeCast = anyValue as? String

    // Let scope function
    val result = nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() {
    // Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function
    fun String.isPalindrome(): Boolean {
        val cleaned = this.lowercase().filter { it.isLetter() }
        return cleaned == cleaned.reversed()
    }

    val sum = add(5, 3)
    val greeting = greet("Kotlin")
    val sumVarargs = sumAll(1, 2, 3, 4, 5)
    val product = operate(4, 5, multiply)
    val palindrome = "Racecar".isPalindrome()
    val fact = factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Basic Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product")
```

```
        Text("Extension: \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact")
    }
}

@Composable
fun demonstrateClassesAndObjects() {
    // Usage examples
    val person = Person("Alice", 25)
    val user = User(1, "john_doe", "john@example.com")
    val priority = Priority.HIGH
    val config = AppConfig.getConfig()
    val area = MathUtils.circleArea(5.0)
    val dog = Dog("Buddy")
    val circle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}")
    }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) {
    var input1 by remember { mutableStateOf("") }
    var input2 by remember { mutableStateOf("") }
    var operation by remember { mutableStateOf("add") }

    Column(verticalArrangement = Arrangement.spacedBy(8.dp)) {
        Text("Try it yourself!")

        OutlinedTextField(
            value = input1,
            onValueChange = { input1 = it },
            label = { Text("Number 1") },
            modifier = Modifier.fillMaxWidth()
        )

        OutlinedTextField(
            value = input2,
            onValueChange = { input2 = it },
            label = { Text("Number 2") },
            modifier = Modifier.fillMaxWidth()
```

```
        )

        Row(
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            modifier = Modifier.fillMaxWidth()
        ) {
            RadioButton(
                selected = operation == "add",
                onClick = { operation = "add" }
            )
            Text("Add")

            RadioButton(
                selected = operation == "subtract",
                onClick = { operation = "subtract" }
            )
            Text("Subtract")

            RadioButton(
                selected = operation == "multiply",
                onClick = { operation = "multiply" }
            )
            Text("Multiply")
        }

        Button(
            onClick = {
                val num1 = input1.toIntOrNull() ?: 0
                val num2 = input2.toIntOrNull() ?: 0
                val result = when (operation) {
                    "add" -> num1 + num2
                    "subtract" -> num1 - num2
                    "multiply" -> num1 * num2
                    else -> 0
                }
                onResult("$num1 $operation $num2 = $result")
            },
            modifier = Modifier.align(Alignment.CenterHorizontally)
        ) {
            Text("Calculate")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
    Basic1Theme {
```

```
        KotlinFundamentalsDemo()
    }
}
```

Output



Expt No : 3 **Date:**

Classes and Objects

Aim

To implement object-oriented programming concepts using classes and objects in Kotlin.

Algorithm

1. Define a class using class keyword.
2. Declare properties inside the class.
3. Define member functions.
4. Create object using constructor.
5. Access class members.
6. Execute and verify output.

Coding

```
package com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview
import com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b
fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!"
fun sumAll(vararg numbers: Int): Int = numbers.sum()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else factorial(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) {
    fun introduce(): String = "Hi, I'm $name and I'm $age years old"
}
```

```
data class User(val id: Int, val username: String, val email: String)

enum class Priority(val value: Int) {
    LOW(1), MEDIUM(2), HIGH(3), URGENT(4);
    fun getDisplayName(): String = name.lowercase().replaceFirstChar { it.uppercase() }
}

object AppConfig {
    const val APP_NAME = "Kotlin Demo"
    const val VERSION = "1.0.0"
    fun getConfig(): String = "$APP_NAME v$VERSION"
}

class MathUtils {
    companion object {
        const val PI = 3.14159
        fun circleArea(radius: Double): Double = PI * radius * radius
    }
}

abstract class Animal(val name: String) {
    abstract fun makeSound(): String
    fun sleep(): String = "$name is sleeping"
}

class Dog(name: String) : Animal(name) {
    override fun makeSound(): String = "$name says: Woof!"
}

interface Drawable {
    fun draw(): String
}

class Circle(private val radius: Double) : Drawable {
    override fun draw(): String = "Drawing a circle with radius $radius"
}

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Basic1Theme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    KotlinFundamentalsDemo(
                        modifier = Modifier.padding(innerPadding)
                    )
```

```
                }
            }
        }
    }
}

@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column(
        modifier = modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text = "Kotlin Fundamentals Demo",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") {
            demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
        }
```

```
        // Classes and Objects
        SectionCard(title = "Classes & Objects") {
            demonstrateClassesAndObjects()
        }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult ->
                result = newResult
            }
        }

        if (result.isNotEmpty()) {
            Card(
                modifier = Modifier.fillMaxWidth(),
                colors = CardDefaults.cardColors(
                    containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text(
                    text = "Result: $result",
                    modifier = Modifier.padding(16.dp),
                    fontSize = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(
        modifier = Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(
            modifier = Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text(
                text = title,
                fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
        }
    }
```

```
}

@Composable
fun demonstrateVariablesAndDataTypes() {
    // Immutable variable (val)
    val immutableName: String = "Kotlin"

    // Mutable variable (var)
    var mutableAge: Int = 10
    mutableAge += 1

    // Type inference
    val inferredDouble = 3.14159

    // Different data types
    val byteValue: Byte = 127
    val shortValue: Short = 32767
    val intValue: Int = 2147483647
    val longValue: Long = 9223372036854775807L
    val floatValue: Float = 3.14f
    val doubleValue: Double = 2.71828
    val charValue: Char = 'K'
    val booleanValue: Boolean = true
    val stringValue: String = "Hello Kotlin"
    val arrayValue: IntArray = intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val (immutable): $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type inference: $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long: $longValue")
        Text("Float: $floatValue")
        Text("Double: $doubleValue")
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}")
    }
}

@Composable
fun demonstrateOperators() {
    val a = 10
    val b = 3

    // Arithmetic operators
```

```
    val sum = a + b
    val difference = a - b
    val product = a * b
    val quotient = a / b
    val remainder = a % b

    // Comparison operators
    val isEqual = a == b
    val isNotEqual = a != b
    val isGreater = a > b
    val isLess = a < b

    // Logical operators
    val x = true
    val y = false
    val andResult = x && y
    val orResult = x || y
    val notResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text("  a + b = $sum")
        Text("  a - b = $difference")
        Text("  a * b = $product")
        Text("  a / b = $quotient")
        Text("  a % b = $remainder")

        Text("Comparison Operators:")
        Text("  a == b = $isEqual")
        Text("  a != b = $isNotEqual")
        Text("  a > b = $isGreater")
        Text("  a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text("  x && y = $andResult")
        Text("  x || y = $orResult")
        Text("  !x = $notResult")
    }
}

@Composable
fun demonstrateControlStatements() {
    val score = 85
    val grade = when {
        score >= 90 -> "A"
        score >= 80 -> "B"
        score >= 70 -> "C"
        score >= 60 -> "D"
```

```
        else -> "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5)
    val evenNumbers = mutableListOf<Int>()

    // For loop
    for (number in numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation
    var countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ")
            countdown--
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text("  Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text("  Numbers: ${numbers.joinToString()}")
        Text("  Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text("  Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    val nullableString: String? = "Hello"
    val nullString: String? = null

    // Safe call operator
    val length1 = nullableString?.length

    // Elvis operator
    val length2 = nullString?.length ?: 0
```

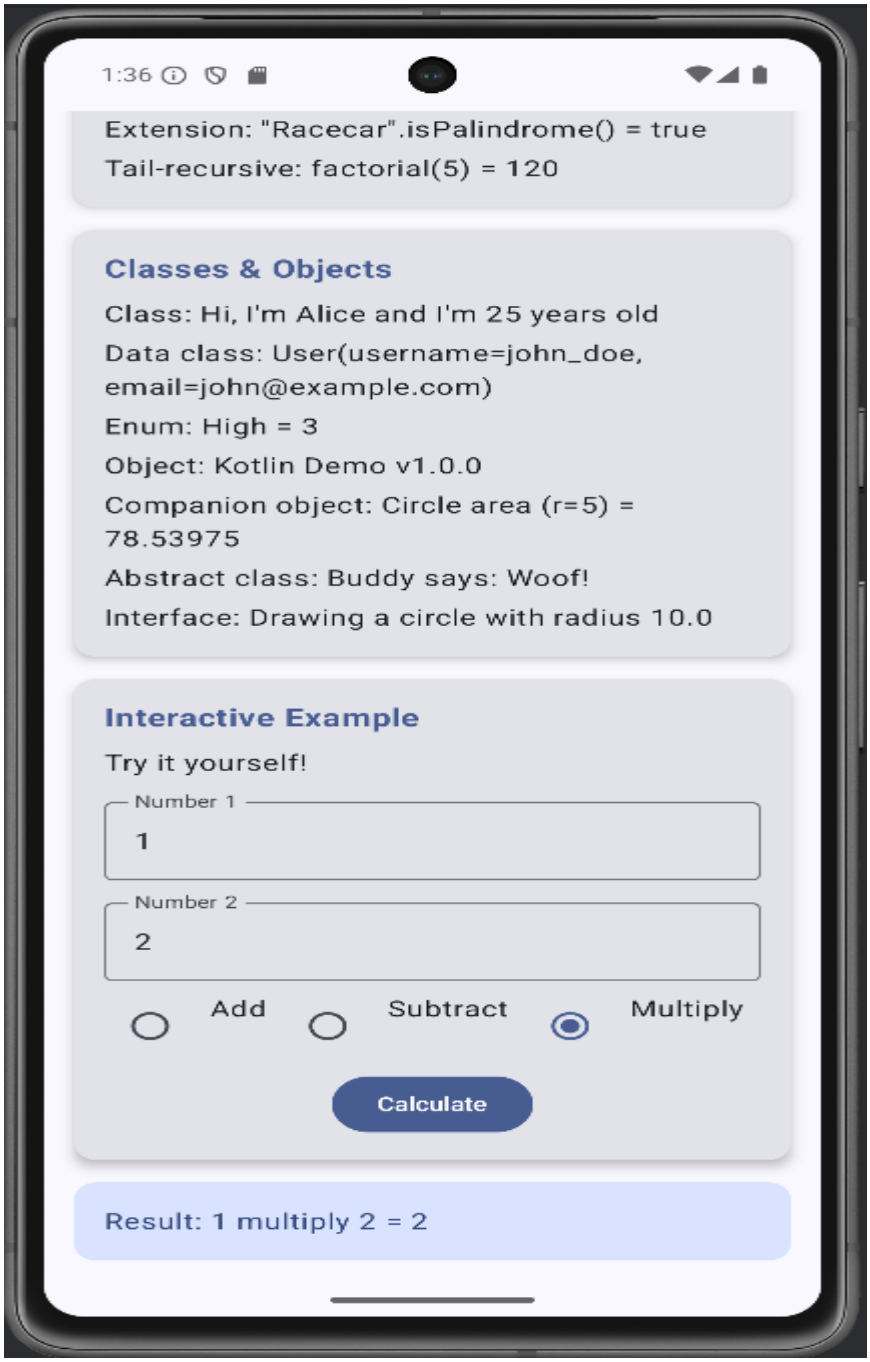
```
    // Non-null asserted call (use with caution)
    val length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin"
    val safeCast = anyValue as? String

    // Let scope function
    val result = nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() {
    // Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function
    fun String.isPalindrome(): Boolean {
        val cleaned = this.lowercase().filter { it.isLetter() }
        return cleaned == cleaned.reversed()
    }

    val sum = add(5, 3)
    val greeting = greet("Kotlin")
    val sumVarargs = sumAll(1, 2, 3, 4, 5)
    val product = operate(4, 5, multiply)
    val palindrome = "Racecar".isPalindrome()
    val fact = factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Basic Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product")
        Text("Extension: \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact")
```

```
    }
}

@Composable
fun demonstrateClassesAndObjects() {
    // Usage examples
    val person = Person("Alice", 25)
    val user = User(1, "john_doe", "john@example.com")
    val priority = Priority.HIGH
    val config = AppConfig.getConfig()
    val area = MathUtils.circleArea(5.0)
    val dog = Dog("Buddy")
    val circle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}")
    }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) {
    var input1 by remember { mutableStateOf("") }
    var input2 by remember { mutableStateOf("") }
    var operation by remember { mutableStateOf("add") }

    Column(verticalArrangement = Arrangement.spacedBy(8.dp)) {
        Text("Try it yourself!")

        OutlinedTextField(
            value = input1,
            onValueChange = { input1 = it },
            label = { Text("Number 1") },
            modifier = Modifier.fillMaxWidth()
        )

        OutlinedTextField(
            value = input2,
            onValueChange = { input2 = it },
            label = { Text("Number 2") },
            modifier = Modifier.fillMaxWidth()
        )
```

```
        Row(
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            modifier = Modifier.fillMaxWidth()
        ) {
            RadioButton(
                selected = operation == "add",
                onClick = { operation = "add" }
            )
            Text("Add")

            RadioButton(
                selected = operation == "subtract",
                onClick = { operation = "subtract" }
            )
            Text("Subtract")

            RadioButton(
                selected = operation == "multiply",
                onClick = { operation = "multiply" }
            )
            Text("Multiply")
        }

        Button(
            onClick = {
                val num1 = input1.toIntOrNull() ?: 0
                val num2 = input2.toIntOrNull() ?: 0
                val result = when (operation) {
                    "add" -> num1 + num2
                    "subtract" -> num1 - num2
                    "multiply" -> num1 * num2
                    else -> 0
                }
                onResult("$num1 $operation $num2 = $result")
            },
            modifier = Modifier.align(Alignment.CenterHorizontally)
        ) {
            Text("Calculate")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
    Basic1Theme {
        KotlinFundamentalsDemo()
    }
```

}

Output


1:36
Extension: "Racecar".isPalindrome() = true
Tail-recursive: factorial(5) = 120
Classes & Objects
Class: Hi, I'm Alice and I'm 25 years old
Data class: User(username=john_doe, email=john@example.com)
Enum: High = 3
Object: Kotlin Demo v1.0.0
Companion object: Circle area (r=5) = 78.53975
Abstract class: Buddy says: Woof!
Interface: Drawing a circle with radius 10.0
Interactive Example
Try it yourself!
Number 1
1
Number 2
2
Add
Subtract
Multiply
Calculate
Result: 1 multiply 2 = 2

Expt No : 4 **Date:**

Build First Android App

Aim

To create and execute a basic Android application using Kotlin.

Algorithm

1. Open Android Studio.
2. Create new project → Empty Activity.
3. Design UI using XML layout.
4. Write Kotlin logic in Activity file.
5. Build the project.
6. Run app on emulator/device.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">
<application
android:allowBackup="true"
android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:supportsRtl="true"
android:theme="@style/Theme.GraphicalPrimitives"
tools:targetApi="31">
<activity
android:name=".MainActivity"
android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<org.rajalakshmi.graphicalprimitives.SampleCanvas
android:layout_width="match_parent"
android:layout_height="match_parent">
</org.rajalakshmi.graphicalprimitives.SampleCanvas>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.kt

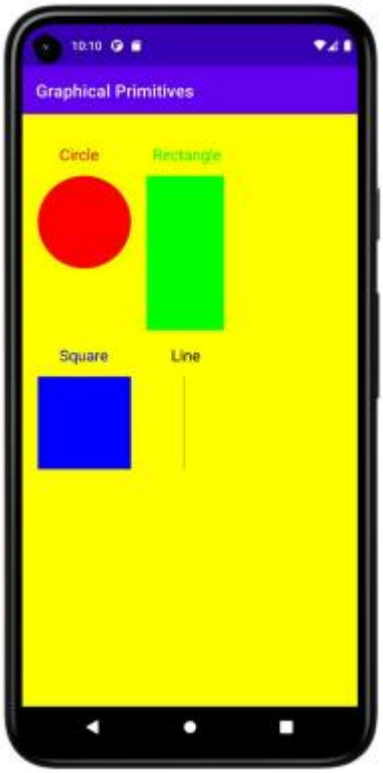
```
package org.rajalakshmi.graphicalprimitives
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
}
}
```

SampleCanvas.kt

```
package org.rajalakshmi.graphicalprimitives
import android.content.Context
import android.graphics.Canvas
import android.graphics.Color
import android.graphics.Paint
import android.util.AttributeSet
import android.view.View
class SampleCanvas @JvmOverloads constructor(
context: Context, attrs: AttributeSet? = null, defStyleAttr: Int = 0
) : View(context, attrs, defStyleAttr) {
override fun onDraw(canvas: Canvas?) {
super.onDraw(canvas)
val paint : Paint = Paint()
paint.setColor(Color.YELLOW)
canvas?.drawPaint(paint)
paint.setTextSize(50f);
paint.setColor(Color.RED);
canvas?.drawText("Circle", 120f, 150f, paint);
canvas?.drawCircle(200f, 350f, 150f, paint);
paint.setColor(Color.GREEN);

canvas?.drawText("Rectangle", 420f, 150f, paint);
canvas?.drawRect(400f, 200f, 650f, 700f, paint);
```

```
paint.setColor(Color.BLUE);
canvas?.drawText("Square", 120f, 800f, paint);
canvas?.drawRect(50f, 850f, 350f, 1150f, paint);
paint.setColor(Color.BLACK);
canvas?.drawText("Line", 480f, 800f, paint);
canvas?.drawLine(520f, 850f, 520f, 1150f, paint);
}
}
```

Output



Expt No : 5 **Date:**

Layouts

Aim

To design user interfaces using Android layout managers.

Algorithm

1. Open XML layout file.
2. Select layout type.
3. Add UI components.
4. Set layout parameters.
5. Preview and run application.

Mainactivity.kt

```
package com.example.androidlayoutmanagers

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.material3.ExperimentalMaterial3Api
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import com.example.androidlayoutmanagers.ui.theme.AndroidLayoutManagersTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            AndroidLayoutManagersTheme {
                Surface(
                    modifier = Modifier.fillMaxSize(),
                    color = MaterialTheme.colorScheme.background
                ) {
                    LayoutManagersApp()
```

```
            }
        }
    }
  }
}

@Composable
fun LayoutManagersApp() {
  var selectedDemo by remember { mutableStateOf<DemoType?>(null) }

  selectedDemo?.let { demo ->
    DemoScreen(demo = demo, onBack = { selectedDemo = null })
  } ?: MainScreen(onDemoSelected = { selectedDemo = it })
}

@Composable
fun MainScreen(onDemoSelected: (DemoType) -> Unit) {
  Column(
    modifier = Modifier
      .fillMaxSize()
      .verticalScroll(rememberScrollState())
      .padding(16.dp),
    verticalArrangement = Arrangement.spacedBy(16.dp)
  ) {
    Text(
      text = "Android Layout Managers",
      fontSize = 32.sp,
      fontWeight = FontWeight.Bold,
      modifier = Modifier.align(Alignment.CenterHorizontally)
    )

    Text(
      text = "Explore different layout managers in Jetpack Compose",
      fontSize = 16.sp,
      color = MaterialTheme.colorScheme.onSurfaceVariant,
      modifier = Modifier.align(Alignment.CenterHorizontally)
    )

    Spacer(modifier = Modifier.height(16.dp))

    DemoType.values().forEach { demo ->
      Card(
        modifier = Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp),
        onClick = { onDemoSelected(demo) }
      ) {
        Column(
          modifier = Modifier
```

```
                .fillMaxWidth()
                .padding(16.dp)
            ) {
                Text(
                    text = demo.title,
                    fontSize = 20.sp,
                    fontWeight = FontWeight.Bold
                )
                Spacer(modifier = Modifier.height(4.dp))
                Text(
                    text = demo.description,
                    fontSize = 14.sp,
                    color = MaterialTheme.colorScheme.onSurfaceVariant
                )
            }
        }
    }
  }
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun DemoScreen(demo: DemoType, onBack: () -> Unit) {
  Column(modifier = Modifier.fillMaxSize()) {
    TopAppBar(
      title = { Text(demo.title) },
      navigationIcon = {
        IconButton(onClick = onBack) {
          Text("← Back")
        }
      }
    )

    Box(modifier = Modifier.fillMaxSize()) {
      when (demo) {
        DemoType.COLUMN -> ColumnExample()
        DemoType.ROW -> RowExample()
        DemoType.BOX -> BoxExample()
        DemoType.LAZY_COLUMN -> LazyColumnExample()
        DemoType.LAZY_ROW -> LazyRowExample()
        DemoType.MIXED_LAZY -> MixedLazyLayoutExample()
        DemoType.CONSTRAINT_BASIC -> ConstraintLayoutBasicExample()
        DemoType.CONSTRAINT_COMPLEX -> ConstraintLayoutComplexExample()
        DemoType.CONSTRAINT_GUIDELINES ->
ConstraintLayoutGuidelinesExample()
        DemoType.GRID_FIXED -> LazyVerticalGridExample()
        DemoType.GRID_ADAPTIVE -> AdaptiveGridExample()
        DemoType.GRID_VARIABLE -> VariableGridExample()
```

```
                DemoType.PRODUCT_GRID -> ProductGridExample()
            }
        }
    }
}

enum class DemoType(
    val title: String,
    val description: String
) {
    COLUMN("Column Layout", "Vertical arrangement of components"),
    ROW("Row Layout", "Horizontal arrangement of components"),
    BOX("Box Layout", "Stacked components with positioning"),
    LAZY_COLUMN("LazyColumn", "Efficient vertical scrolling lists"),
    LAZY_ROW("LazyRow", "Efficient horizontal scrolling lists"),
    MIXED_LAZY("Mixed Lazy Layout", "Combination of LazyColumn and LazyRow"),
    CONSTRAINT_BASIC("ConstraintLayout Basic", "Basic constraint positioning"),
    CONSTRAINT_COMPLEX("ConstraintLayout Complex", "Complex profile layout"),
    CONSTRAINT_GUIDELINES("ConstraintLayout Guidelines", "Layout with
guidelines"),
    GRID_FIXED("Fixed Grid", "Fixed number of columns"),
    GRID_ADAPTIVE("Adaptive Grid", "Adaptive column sizing"),
    GRID_VARIABLE("Variable Grid", "Variable item sizes"),
    PRODUCT_GRID("Product Grid", "Real-world product grid")
}
```

settings.gradle.kts

```
pluginManagement {
    repositories {
        google {
            content {
                includeGroupByRegex("com\\.android.*")
                includeGroupByRegex("com\\.google.*")
                includeGroupByRegex("androidx.*")
            }
        }
        mavenCentral()
        gradlePluginPortal()
    }
}
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
    repositories {
        google()
        mavenCentral()
```

```
    }
}

rootProject.name = "Android layout managers"
include(":app")
```

ConstraintsLayoutExamples.kts

```
package com.example.androidlayoutmanagers
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.constraintlayout.compose.ConstraintLayout
import androidx.constraintlayout.compose.ConstraintSet
import androidx.constraintlayout.compose.Dimension

@Composable
fun ConstraintLayoutBasicExample() {
    ConstraintLayout(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ) {
        val (title, subtitle, button, image) = createRefs()

        Text(
            text = "ConstraintLayout Basic",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.constrainAs(title) {
                top.linkTo(parent.top, margin = 16.dp)
```

```
            start.linkTo(parent.start)
            end.linkTo(parent.end)
        }
    )

    Text(
        text = "Positioning elements with constraints",
        fontSize = 16.sp,
        color = MaterialTheme.colorScheme.onSurfaceVariant,
        modifier = Modifier.constrainAs(subtitle) {
            top.linkTo(title.bottom, margin = 8.dp)
            start.linkTo(parent.start)
            end.linkTo(parent.end)
        }
    )

    Button(
        onClick = { },
        modifier = Modifier.constrainAs(button) {
            bottom.linkTo(parent.bottom, margin = 32.dp)
            start.linkTo(parent.start)
            end.linkTo(parent.end)
        }
    ) {
        Text("Centered Button")
    }

    Box(
        modifier = Modifier
            .size(100.dp)
            .background(
                Color.Blue,
                RoundedCornerShape(12.dp)
            )
            .constrainAs(image) {
                top.linkTo(subtitle.bottom, margin = 24.dp)
                bottom.linkTo(button.top, margin = 24.dp)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            },
        contentAlignment = Alignment.Center
    ) {
        Text(
            text = "Center Box",
            color = Color.White,
            fontWeight = FontWeight.Bold
        )
    }
```

```
    }
}

@Composable
fun ConstraintLayoutComplexExample() {
    ConstraintLayout(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ) {
        val (header, profilePic, userName, userEmail, statsContainer,
             bioText, actionButton, settingsButton) = createRefs()

        val barrier = createEndBarrier(profilePic, userName, userEmail)

        Text(
            text = "Profile Layout",
            fontSize = 20.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.constrainAs(header) {
                top.linkTo(parent.top)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            }
        )

        Box(
            modifier = Modifier
                .size(80.dp)
                .background(
                    Color.Green,
                    RoundedCornerShape(40.dp)
                )
                .constrainAs(profilePic) {
                    top.linkTo(header.bottom, margin = 16.dp)
                    start.linkTo(parent.start)
                },
            contentAlignment = Alignment.Center
        ) {
            Text(
                text = "PIC",
                color = Color.White,
                fontWeight = FontWeight.Bold
            )
        }

        Text(
            text = "John Doe",
```

```
            fontSize = 18.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.constrainAs(userName) {
                start.linkTo(profilePic.end, margin = 16.dp)
                top.linkTo(profilePic.top)
            }
        )

        Text(
            text = "john.doe@example.com",
            fontSize = 14.sp,
            color = MaterialTheme.colorScheme.onSurfaceVariant,
            modifier = Modifier.constrainAs(userEmail) {
                start.linkTo(profilePic.end, margin = 16.dp)
                top.linkTo(userName.bottom, margin = 4.dp)
            }
        )

        Box(
            modifier = Modifier
                .constrainAs(statsContainer) {
                    top.linkTo(profilePic.bottom, margin = 16.dp)
                    start.linkTo(parent.start)
                    end.linkTo(parent.end)
                    width = Dimension.fillToConstraints
                }
        ) {
            Row(
                modifier = Modifier.fillMaxWidth(),
                horizontalArrangement = Arrangement.SpaceEvenly
            ) {
                StatItem("Posts", "150", Modifier)
                StatItem("Followers", "2.5K", Modifier)
                StatItem("Following", "500", Modifier)
            }
        }

        Text(
            text = "Passionate developer and Android enthusiast. Love creating beautiful and
functional mobile applications.",
            fontSize = 14.sp,
            modifier = Modifier.constrainAs(bioText) {
                top.linkTo(statsContainer.bottom, margin = 16.dp)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
                width = Dimension.fillToConstraints
            }
        )
```

```
        Button(
            onClick = { },
            modifier = Modifier.constrainAs(actionButton) {
                top.linkTo(bioText.bottom, margin = 16.dp)
                start.linkTo(parent.start)
                end.linkTo(settingsButton.start, margin = 8.dp)
                width = Dimension.fillToConstraints
            }
        ) {
            Text("Follow")
        }

        OutlinedButton(
            onClick = { },
            modifier = Modifier.constrainAs(settingsButton) {
                top.linkTo(bioText.bottom, margin = 16.dp)
                end.linkTo(parent.end)
                width = Dimension.value(120.dp)
            }
        ) {
            Text("Message")
        }
    }
}

@Composable
private fun StatItem(label: String, value: String, modifier: Modifier) {
    Column(
        horizontalAlignment = Alignment.CenterHorizontally,
        modifier = modifier
    ) {
        Text(
            text = value,
            fontSize = 18.sp,
            fontWeight = FontWeight.Bold
        )
        Text(
            text = label,
            fontSize = 12.sp,
            color = MaterialTheme.colorScheme.onSurfaceVariant
        )
    }
}

@Composable
fun ConstraintLayoutGuidelinesExample() {
    ConstraintLayout(
```

```
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ) {
        val (title, leftCard, centerCard, rightCard, bottomBar) = createRefs()

        val verticalGuideline = createGuidelineFromStart(0.33f)
        val verticalGuideline2 = createGuidelineFromEnd(0.33f)
        val horizontalGuideline = createGuidelineFromTop(0.4f)

        Text(
            text = "Guidelines Layout",
            fontSize = 20.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.constrainAs(title) {
                top.linkTo(parent.top)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            }
        )

        Card(
            modifier = Modifier
                .fillMaxHeight()
                .constrainAs(leftCard) {
                    top.linkTo(title.bottom, margin = 16.dp)
                    bottom.linkTo(bottomBar.top, margin = 16.dp)
                    start.linkTo(parent.start)
                    end.linkTo(verticalGuideline)
                },
            colors = CardDefaults.cardColors(
                containerColor = Color.Red.copy(alpha = 0.7f)
            )
        ) {
            Box(
                contentAlignment = Alignment.Center,
                modifier = Modifier.fillMaxSize()
            ) {
                Text(
                    text = "Left\n33%",
                    color = Color.White,
                    fontWeight = FontWeight.Bold
                )
            }
        }

        Card(
            modifier = Modifier
```

```
            .fillMaxHeight()
            .constrainAs(centerCard) {
                top.linkTo(title.bottom, margin = 16.dp)
                bottom.linkTo(bottomBar.top, margin = 16.dp)
                start.linkTo(verticalGuideline)
                end.linkTo(verticalGuideline2)
            },
        colors = CardDefaults.cardColors(
            containerColor = Color.Green.copy(alpha = 0.7f)
        )
    ) {
        Box(
            contentAlignment = Alignment.Center,
            modifier = Modifier.fillMaxSize()
        ) {
            Text(
                text = "Center\n34%",
                color = Color.White,
                fontWeight = FontWeight.Bold
            )
        }
    }

    Card(
        modifier = Modifier
            .fillMaxHeight()
            .constrainAs(rightCard) {
                top.linkTo(title.bottom, margin = 16.dp)
                bottom.linkTo(bottomBar.top, margin = 16.dp)
                start.linkTo(verticalGuideline2)
                end.linkTo(parent.end)
            },
        colors = CardDefaults.cardColors(
            containerColor = Color.Blue.copy(alpha = 0.7f)
        )
    ) {
        Box(
            contentAlignment = Alignment.Center,
            modifier = Modifier.fillMaxSize()
        ) {
            Text(
                text = "Right\n33%",
                color = Color.White,
                fontWeight = FontWeight.Bold
            )
        }
    }
```

```
        Card(
            modifier = Modifier
                .height(60.dp)
                .constrainAs(bottomBar) {
                    bottom.linkTo(parent.bottom)
                    start.linkTo(parent.start)
                    end.linkTo(parent.end)
                },
            colors = CardDefaults.cardColors(
                containerColor = MaterialTheme.colorScheme.surfaceVariant
            )
        ) {
            Box(
                contentAlignment = Alignment.Center,
                modifier = Modifier.fillMaxSize()
            ) {
                Text(
                    text = "Bottom Navigation Bar",
                    fontWeight = FontWeight.Medium
                )
            }
        }
    }
}
```

GridLayoutExamples.kt

```
package com.example.androidlayoutmanagers
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.grid.GridCells
import androidx.compose.foundation.lazy.grid.LazyVerticalGrid
import androidx.compose.foundation.lazy.grid.items
import androidx.compose.foundation.lazy.grid.rememberLazyGridState
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp

data class GridItem(val id: Int, val title: String, val color: Color)

@Composable
```

```
fun LazyVerticalGridExample() {
    val items = remember {
        (1..50).map { index ->
            val colors = listOf(
                Color.Red, Color.Blue, Color.Green, Color.Yellow,
                Color.Magenta, Color.Cyan, Color.Gray, Color(0xFFFF6B35)
            )
            GridItem(
                id = index,
                title = "Item $index",
                color = colors[index % colors.size]
            )
        }
    }

    val gridState = rememberLazyGridState()

    Column(
        modifier = Modifier.fillMaxSize()
    ) {
        Text(
            text = "LazyVerticalGrid - 2 Columns",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.padding(16.dp)
        )

        LazyVerticalGrid(
            columns = GridCells.Fixed(2),
            modifier = Modifier.fillMaxSize(),
            state = gridState,
            contentPadding = PaddingValues(16.dp),
            horizontalArrangement = Arrangement.spacedBy(12.dp),
            verticalArrangement = Arrangement.spacedBy(12.dp)
        ) {
            items(items) { item ->
                Card(
                    modifier = Modifier
                        .fillMaxWidth()
                        .aspectRatio(1f),
                    elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
                ) {
                    Box(
                        modifier = Modifier
                            .fillMaxSize()
                            .background(item.color.copy(alpha = 0.8f)),
                        contentAlignment = Alignment.Center
                    ) {
```

```
                    Text(
                        text = item.title,
                        color = Color.White,
                        fontWeight = FontWeight.Bold,
                        fontSize = 16.sp
                    )
                }
            }
        }
    }
}

@Composable
fun AdaptiveGridExample() {
    val items = remember {
        (1..30).map { index ->
            val colors = listOf(
                Color(0xFF6200EE), Color(0xFF03DAC6), Color(0xFFCF6679),
                Color(0xFF3700B3), Color(0xFF018786), Color(0xFFBB86FC)
            )
            GridItem(
                id = index,
                title = "Adaptive $index",
                color = colors[index % colors.size]
            )
        }
    }

    Column(
        modifier = Modifier.fillMaxSize()
    ) {
        Text(
            text = "Adaptive Grid - Min 120dp",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.padding(16.dp)
        )

        LazyVerticalGrid(
            columns = GridCells.Adaptive(minSize = 120.dp),
            modifier = Modifier.fillMaxSize(),
            contentPadding = PaddingValues(16.dp),
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            items(items) { item ->
                Card(
```

```
                modifier = Modifier
                    .fillMaxWidth()
                    .height(100.dp),
                elevation = CardDefaults.cardElevation(defaultElevation = 3.dp)
            ) {
                Box(
```

LazyLayoutExamples.kt

```
package com.example.androidlayoutmanagers

import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.LazyRow
import androidx.compose.foundation.lazy.items
import androidx.compose.foundation.lazy.rememberLazyListState
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp

data class Item(val id: Int, val title: String, val description: String)

@Composable
fun LazyColumnExample() {
    val items = remember {
        (1..50).map { index ->
            Item(
                id = index,
                title = "Item $index",
                description = "This is the description for item number $index"
            )
        }
    }

    val listState = rememberLazyListState()

    Column(
        modifier = Modifier.fillMaxSize()
```

```
    ) {
        Text(
            text = "LazyColumn - ${items.size} items",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.padding(16.dp)
        )

        LazyColumn(
            modifier = Modifier.fillMaxSize(),
            state = listState,
            contentPadding = PaddingValues(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            items(items) { item ->
                Card(
                    modifier = Modifier
                        .fillMaxWidth(),
                    elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
                ) {
                    Column(
                        modifier = Modifier
                            .fillMaxWidth()
                            .padding(16.dp)
                    ) {
                        Text(
                            text = item.title,
                            fontSize = 18.sp,
                            fontWeight = FontWeight.Bold,
                            color = MaterialTheme.colorScheme.primary
                        )
                        Spacer(modifier = Modifier.height(4.dp))
                        Text(
                            text = item.description,
                            fontSize = 14.sp,
                            color = MaterialTheme.colorScheme.onSurfaceVariant
                        )
                    }
                }
            }
        }
    }
}

@Composable
fun LazyRowExample() {
    val categories = remember {
        listOf(
```

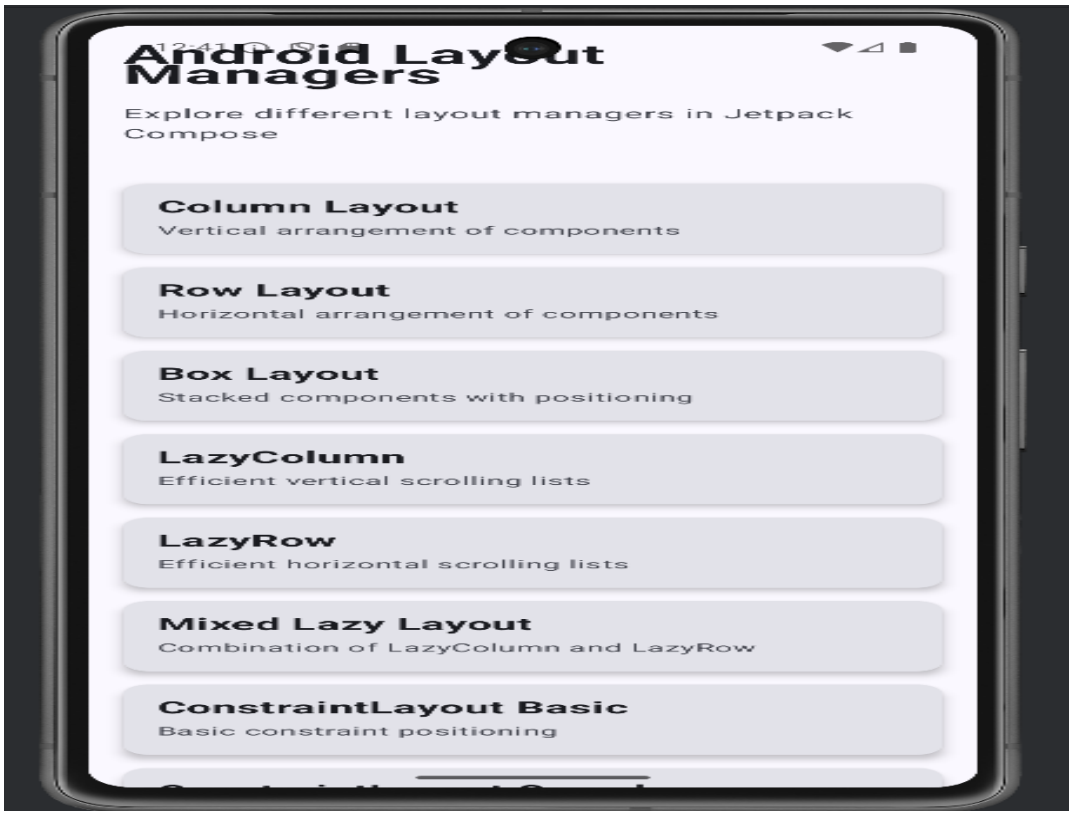
```
        "Technology", "Sports", "Music", "Movies", "Books",
        "Travel", "Food", "Fashion", "Gaming", "Art",
        "Science", "History", "Nature", "Photography", "Dance"
    )
}

val listState = rememberLazyListState()

Column(
    modifier = Modifier
        .fillMaxSize()
        .padding(16.dp)
) {
    Text(
        text = "LazyRow - Horizontal Scrolling",
        fontSize = 24.sp,
        fontWeight = FontWeight.Bold,
        modifier = Modifier.padding(bottom = 16.dp)
    )

    LazyRow(
        state = listState,
        horizontalArrangement = Arrangement.spacedBy(12.dp),
        contentPadding = PaddingValues(horizontal = 16.dp)
    ) {
        items(categories) { category ->
            Card(
                modifier = Modifier
                    .width(120.dp)
                    .height(80.dp),
                shape = RoundedCornerShape(12.dp),
                elevation = CardDefaults.cardElevation(defaultElevation = 6.dp)
            ) {
                Box(
                    modifier = Modifier
                        .fillMaxSize()
                        .background(
                            Color(
                                red = (0.3f + (category.hashCode() % 100) / 200f),
                                green = (0.5f + (category.hashCode() % 150) / 300f),
                                blue = (0.7f + (category.hashCode() % 200) / 400f)
                            )
                        ),
                    contentAlignment = Alignment.Center
                ) {
                    Text(
                        text = category,
                        color = Color.White,
```

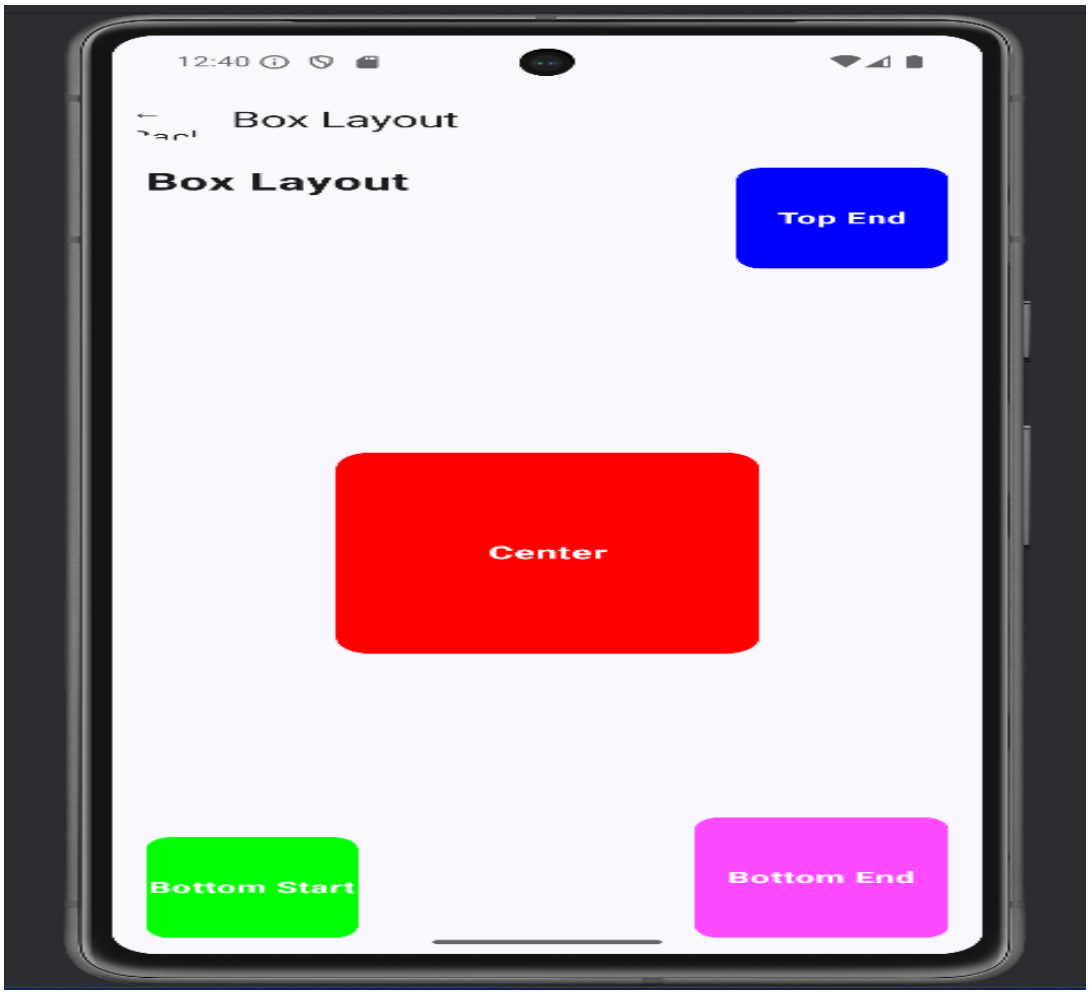
```
                fontWeight = FontWeight.Bold,
                modifier = Modifier.padding(8.dp)
            )
        }
    }
  }
}

Spacer(modifier = Modifier.height(24.dp))

Text(
  text = "Featured Items",
  fontSize = 20.sp,
  fontWeight = FontWeight.Bold,
  modifier = Modifier.padding(bottom = 12.dp)
)

val featuredItems = remember {
  (1..20).map { index ->
    "Featured $index"
  }
}

LazyRow(
  horizontalArrangement = Arrangement.spacedBy(8.dp),
  contentPadding = PaddingValues(horizontal = 16.dp)
) {
  items(featuredItems) { item ->
    Card(
      modifier = Modifier
        .width(150.dp)
        .height(100.dp),
      colors = CardDefaults.cardColors(
        containerColor = MaterialTheme.colorScheme.surfaceVariant
      )
    ) {
      Box(
        contentAlignment = Alignment.Center
      ) {
        Text(
          text = item,
          fontWeight = FontWeight.Medium,
          modifier = Modifier.padding(12.dp)
        )
      }
    }
  }
}
```

```
    }
}

@Composable
fun MixedLazyLayoutExample() {
    val sections = remember {
        listOf(
            "Header" to listOf("Welcome to Lazy Layouts"),
            "Categories" to listOf("Tech", "Sports", "Music", "Movies", "Books"),
            "Popular Items" to (1..10).map { "Popular Item $it" },
            "Recent Items" to (1..15).map { "Recent Item $it" }
        )
    }

    LazyColumn(
        modifier = Modifier.fillMaxSize(),
        contentPadding = PaddingValues(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        sections.forEach { (sectionTitle, items) ->
            item {
                Text(
                    text = sectionTitle,
                    fontSize = 20.sp,
                    fontWeight = FontWeight.Bold,
                    modifier = Modifier.padding(vertical = 8.dp)
                )
            }

            if (sectionTitle == "Categories") {
                item {
                    LazyRow(
                        horizontalArrangement = Arrangement.spacedBy(8.dp)
                    ) {
                        items(items) { category ->
                            Card(
                                modifier = Modifier
                                    .width(100.dp)
                                    .height(60.dp),
                                colors = CardDefaults.cardColors(
                                    containerColor = MaterialTheme.colorScheme.primaryContainer
                                )
                            ) {
                                Box(
                                    contentAlignment = Alignment.Center
                                ) {
                                    Text(
                                        text = category,
```

```
                                fontWeight = FontWeight.Medium,
                                color = MaterialTheme.colorScheme.onPrimaryContainer
                            )
                        }
                    }
                }
            }
        }
    } else {
        items(items) { item ->
            Card(
                modifier = Modifier.fillMaxWidth(),
                elevation = CardDefaults.cardElevation(defaultElevation = 2.dp)
            ) {
                Text(
                    text = item,
                    modifier = Modifier.padding(16.dp)
                )
            }
        }
    }
  }
}
```

Output




12:40
Box Layout
Box Layout
Top End
Center
Bottom Start
Bottom End

Expt No : 6 **Date:**

App Navigation

Aim

To implement navigation between multiple screens in Android app.

Algorithm

1. Create multiple activities/fragments.
2. Use Intent to navigate.
3. Pass data using extras.
4. Implement Navigation component.
5. Test navigation flow.

CODING

MainActivity.kt

```
package com.example.sai1
import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button

class MainActivity : AppCompatActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)

      val btnPage1: Button = findViewById(R.id.btnPage1)
      val btnPage2: Button = findViewById(R.id.btnPage2)
      val btnPage3: Button = findViewById(R.id.btnPage3)
      val btnPage4: Button = findViewById(R.id.btnPage4)

      btnPage1.setOnClickListener {
         startActivity(Intent(this, Page1Activity::class.java))
      }

      btnPage2.setOnClickListener {
         startActivity(Intent(this, Page2Activity::class.java))
      }

      btnPage3.setOnClickListener {
         startActivity(Intent(this, Page3Activity::class.java))
      }
```

```
        btnPage4.setOnClickListener {
            startActivity(Intent(this, Page4Activity::class.java))
        }
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    android:background="@color/white">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="GUI Components"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp" />

    <Button
        android:id="@+id/btnPage1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 1 - Initial State"
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnPage2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 2 - Large Font"
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnPage3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 3 - Green Text"
```

```
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnPage4"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 4 - Blue Background"
        android:textSize="16sp" />

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportsRtl="true"
        android:theme="@style/Theme.Sai1"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".Page1Activity"
            android:exported="false" />
        <activity
            android:name=".Page2Activity"
            android:exported="false" />
        <activity
            android:name=".Page3Activity"
            android:exported="false" />
        <activity
            android:name=".Page4Activity"
            android:exported="false" />
    </application>
```

</manifest>

OUTPUT

Expt No : 7 **Date:**

Activity and Fragment Lifecycles

Aim

To understand lifecycle methods of Activity and Fragment.

Algorithm

1. Create activity/fragment.
2. Override lifecycle methods.
3. Use Logcat for logging.
4. Rotate device and observe lifecycle.
5. Analyze lifecycle behavior.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
tools:context=".MainActivity">
<TextView
android:id="@+id/tvTitle"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Student Details"
android:textAlignment="center"
android:textSize="24sp" />
<LinearLayout
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical">
<fragment
android:id="@+id/fragmentBasic"
android:name="org.rajalakshmi.androidfragments.StudentBasicDetails"
android:layout_width="match_parent"
android:layout_height="300dp" />
<fragment
android:id="@+id/fragmentMark"

android:name="org.rajalakshmi.androidfragments.StudentMarkDetails"
android:layout_width="match_parent"
android:layout_height="300dp" />
</LinearLayout>
```

```
</LinearLayout>
```

fragment_student_basic_details.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".StudentBasicDetails">
<TextView
android:id="@+id/tvBasicDetails"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Basic Details"
android:textAlignment="center"
android:textSize="24sp" />
<TextView
android:id="@+id/tvRegisterNumber"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="Register No." />
<EditText
android:id="@+id/etRegisterNumber"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:ems="10"
android:hint="Register Number"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvName"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="125dp"
android:text="Name" />
<EditText
android:id="@+id/etName"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="125dp"
android:ems="10"
android:hint="Name"
android:inputType="textPersonName" />
```

```
<TextView
android:id="@+id/tvDepartment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="200dp"
android:text="Department" />
<EditText
android:id="@+id/etDepartment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="200dp"
android:ems="10"
android:hint="Department"
android:inputType="textPersonName" />
</FrameLayout>
```

fragment_student_mark_details.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".StudentMarkDetails">
<TextView
android:id="@+id/tvBasicDetails"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Mark Details"
android:textAlignment="center"
android:textSize="24sp" />
<TextView
android:id="@+id/tvSSLC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="S.S.L.C." />
<EditText
android:id="@+id/etSSLC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:ems="10"
android:hint="S.S.L.C. Mark"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvHSc"
```

```
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="125dp"
android:text="H.Sc." />
<EditText
android:id="@+id/etHSC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="125dp"
android:ems="10"
android:hint="H.Sc. Mark"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvUG"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="200dp"
android:text="U.G." />
<EditText
android:id="@+id/etUG"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="200dp"
android:ems="10"
android:hint="U.G. C.G.P.A."
android:inputType="textPersonName" />
</FrameLayout>
```

MainActivity.kt

```
package org.rajalakshmi.androidfragments
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
}
}
```


StudentBasicDetails.kt

```
package org.rajalakshmi.androidfragments
import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
```

```
import android.view.ViewGroup
// TODO: Rename parameter arguments, choose names that match
// the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private const val ARG_PARAM1 = "param1"
private const val ARG_PARAM2 = "param2"
/**
* A simple [Fragment] subclass.
* Use the [StudentBasicDetails.newInstance] factory method to
* create an instance of this fragment.
*/
class StudentBasicDetails : Fragment() {
// TODO: Rename and change types of parameters
private var param1: String? = null
private var param2: String? = null
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
arguments?.let {
param1 = it.getString(ARG_PARAM1)
param2 = it.getString(ARG_PARAM2)
}
}
override fun onCreateView(
inflater: LayoutInflater, container: ViewGroup?,
savedInstanceState: Bundle?
): View? {
// Inflate the layout for this fragment
return inflater.inflate(R.layout.fragment_student_basic_details,
container, false)
}
companion object {
/**
* Use this factory method to create a new instance of
* this fragment using the provided parameters.
*
* @param param1 Parameter 1.
* @param param2 Parameter 2.
* @return A new instance of fragment StudentBasicDetails.
*/
// TODO: Rename and change types and number of parameters
@JvmStatic
fun newInstance(param1: String, param2: String) =
StudentBasicDetails().apply {
arguments = Bundle().apply {
putString(ARG_PARAM1, param1)
putString(ARG_PARAM2, param2)
}
}
}
```

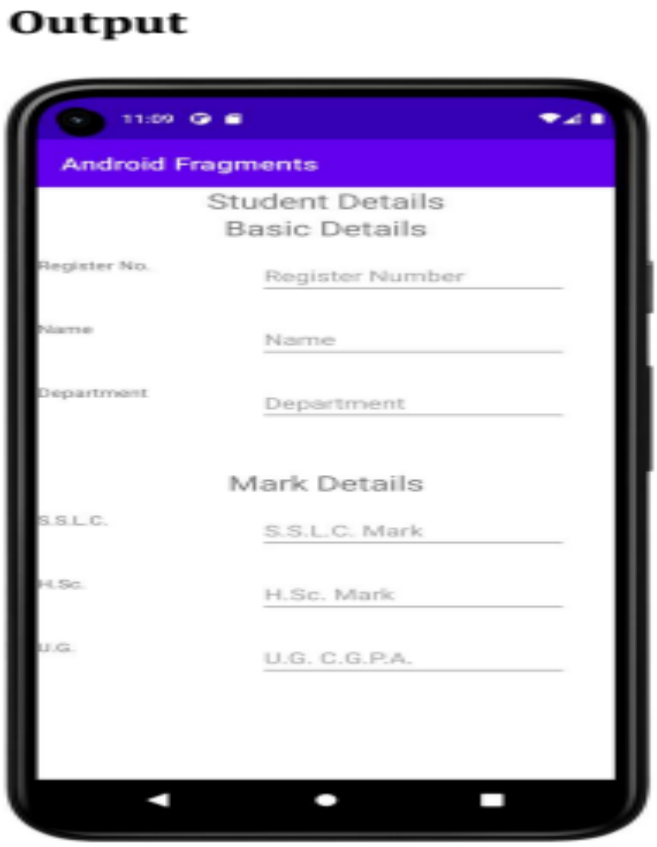
```
}
```


StudentMarkDetails.kt

```
package org.rajalakshmi.androidfragments
import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
// TODO: Rename parameter arguments, choose names that match
// the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private const val ARG_PARAM1 = "param1"
private const val ARG_PARAM2 = "param2"
/**
* A simple [Fragment] subclass.
* Use the [StudentMarkDetails.newInstance] factory method to
* create an instance of this fragment.
*/
class StudentMarkDetails : Fragment() {
// TODO: Rename and change types of parameters
private var param1: String? = null
private var param2: String? = null
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
arguments?.let {
param1 = it.getString(ARG_PARAM1)
param2 = it.getString(ARG_PARAM2)
}
}
override fun onCreateView(
inflater: LayoutInflater, container: ViewGroup?,
savedInstanceState: Bundle?
): View? {
// Inflate the layout for this fragment
return inflater.inflate(R.layout.fragment_student_mark_details,
container, false)
}
companion object {
/**
* Use this factory method to create a new instance of
* this fragment using the provided parameters.
*
* @param param1 Parameter 1.
* @param param2 Parameter 2.
* @return A new instance of fragment StudentMarkDetails.
*/
// TODO: Rename and change types and number of parameters
```

```
@JvmStatic
fun newInstance(param1: String, param2: String) =
StudentMarkDetails().apply {
arguments = Bundle().apply {
putString(ARG_PARAM1, param1)
putString(ARG_PARAM2, param2)
}
}
}
}
```

Output



Expt No : 8 **Date:**

App Architecture (UI Layer)

Aim

To implement UI layer using MVVM architecture.

Algorithm

1. Create ViewModel class.
2. Add LiveData objects.
3. Bind UI with ViewModel.
4. Observe data changes.
5. Update UI automatically.

Coding

Mainactivity.kt

```
package com.example.apparchitectureuilayer

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import com.example.apparchitectureuilayer.presentation.navigation.AppNavigationSimple
import com.example.apparchitectureuilayer.ui.theme.AppArchitectureUILayerTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            AppArchitectureUILayerTheme {
                AppNavigationSimple()
            }
        }
    }
}
```

UserDetailScreen.kt

```
package com.example.apparchitectureuilayer.presentation.screen

import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.shape.CircleShape
import androidx.compose.material3.*
```

```
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.draw.clip
import androidx.compose.ui.layout.ContentScale
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.hilt.navigation.compose.hiltViewModel
import coil.compose.AsyncImage
import com.example.apparchitectureuilayer.data.model.ApiResponse
import
com.example.apparchitectureuilayer.presentation.viewmodel.UserDetailViewModel

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun UserDetailScreen(
    userId: String,
    onBack: () -> Unit,
    viewModel: UserDetailViewModel = hiltViewModel()
) {
    LaunchedEffect(userId) {
        viewModel.loadUser(userId)
    }

    val user by viewModel.user.collectAsState()

    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("User Details") },
                navigationIcon = {
                    IconButton(onClick = onBack) {
                        Text("←")
                    }
                }
            )
        }
    ) { padding ->
        Box(
            modifier = Modifier
                .fillMaxSize()
                .padding(padding)
        ) {
            when (user) {
                is ApiResponse.Loading -> {
                    CircularProgressIndicator(
                        modifier = Modifier.align(Alignment.Center)
                    )
```

```
            }
            is ApiResponse.Success -> {
                val userData = (user as ApiResponse.Success).data
                LazyColumn(
                    modifier = Modifier.fillMaxSize(),
                    contentPadding = PaddingValues(16.dp),
                    verticalArrangement = Arrangement.spacedBy(16.dp)
                ) {
                    item {
                        Row(
                            modifier = Modifier.fillMaxWidth(),
                            horizontalArrangement = Arrangement.Center
                        ) {
                            if (userData.avatar != null) {
                                AsyncImage(
                                    model = userData.avatar,
                                    contentDescription = "User Avatar",
                                    modifier = Modifier
                                        .size(120.dp)
                                        .clip(CircleShape),
                                    contentScale = ContentScale.Crop
                                )
                            } else {
                                Box(
                                    modifier = Modifier
                                        .size(120.dp)
                                        .clip(CircleShape)
                                        .background(MaterialTheme.colorScheme.primaryContainer),
                                    contentAlignment = Alignment.Center
                                ) {
                                    Text(
                                        text = userData.name.first().uppercase(),
                                        style = MaterialTheme.typography.headlineLarge,
                                        color = MaterialTheme.colorScheme.onPrimaryContainer
                                    )
                                }
                            }
                        }
                    }

                    item {
                        Column(
                            horizontalAlignment = Alignment.CenterHorizontally
                        ) {
                            Text(
                                text = userData.name,
                                style = MaterialTheme.typography.headlineMedium,
                                fontWeight = FontWeight.Bold
```

```
            )
            Spacer(modifier = Modifier.height(8.dp))
            Text(
                text = userData.email,
                style = MaterialTheme.typography.bodyLarge,
                color = MaterialTheme.colorScheme.onSurfaceVariant
            )
            Spacer(modifier = Modifier.height(8.dp))
            Text(
                text = if (userData.isActive) "Active" else "Inactive",
                style = MaterialTheme.typography.bodyMedium,
                color = if (userData.isActive)
                    MaterialTheme.colorScheme.primary else
                    MaterialTheme.colorScheme.error,
                modifier = Modifier
                    .padding(horizontal = 16.dp)
                    .padding(vertical = 4.dp)
                    .background(
                        color = if (userData.isActive)
                            MaterialTheme.colorScheme.primaryContainer else
                            MaterialTheme.colorScheme.errorContainer,
                        shape = MaterialTheme.shapes.small
                    )
                    .padding(horizontal = 12.dp, vertical = 4.dp)
            )
        }
    }

    item {
        Card(
            modifier = Modifier.fillMaxWidth()
        ) {
            Column(
                modifier = Modifier
                    .fillMaxWidth()
                    .padding(16.dp)
            ) {
                Text(
                    text = "User Information",
                    style = MaterialTheme.typography.titleMedium,
                    fontWeight = FontWeight.Bold
                )
                Spacer(modifier = Modifier.height(12.dp))

                InfoRow("ID", userData.id)
                InfoRow("Name", userData.name)
                InfoRow("Email", userData.email)
                InfoRow("Status", if (userData.isActive) "Active" else "Inactive")
```

```
                    }
                }
            }
        }
    }
    is ApiResponse.Error -> {
        Column(
            modifier = Modifier.align(Alignment.Center),
            horizontalAlignment = Alignment.CenterHorizontally
        ) {
            Text(
                text = "Error: ${(user as ApiResponse.Error).message}",
                color = MaterialTheme.colorScheme.error
            )
            Spacer(modifier = Modifier.height(8.dp))
            Button(onClick = { viewModel.loadUser(userId) }) {
                Text("Retry")
            }
        }
    }
  }
 }
}

@Composable
private fun InfoRow(label: String, value: String) {
    Column {
        Text(
            text = label,
            style = MaterialTheme.typography.bodySmall,
            color = MaterialTheme.colorScheme.onSurfaceVariant
        )
        Text(
            text = value,
            style = MaterialTheme.typography.bodyMedium,
            modifier = Modifier.padding(top = 2.dp)
        )
        Spacer(modifier = Modifier.height(8.dp))
    }
}
```

AppNavigationSimple.kt

```
package com.example.apparchitectureuilayer.presentation.navigation
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.RoundedCornerShape
```

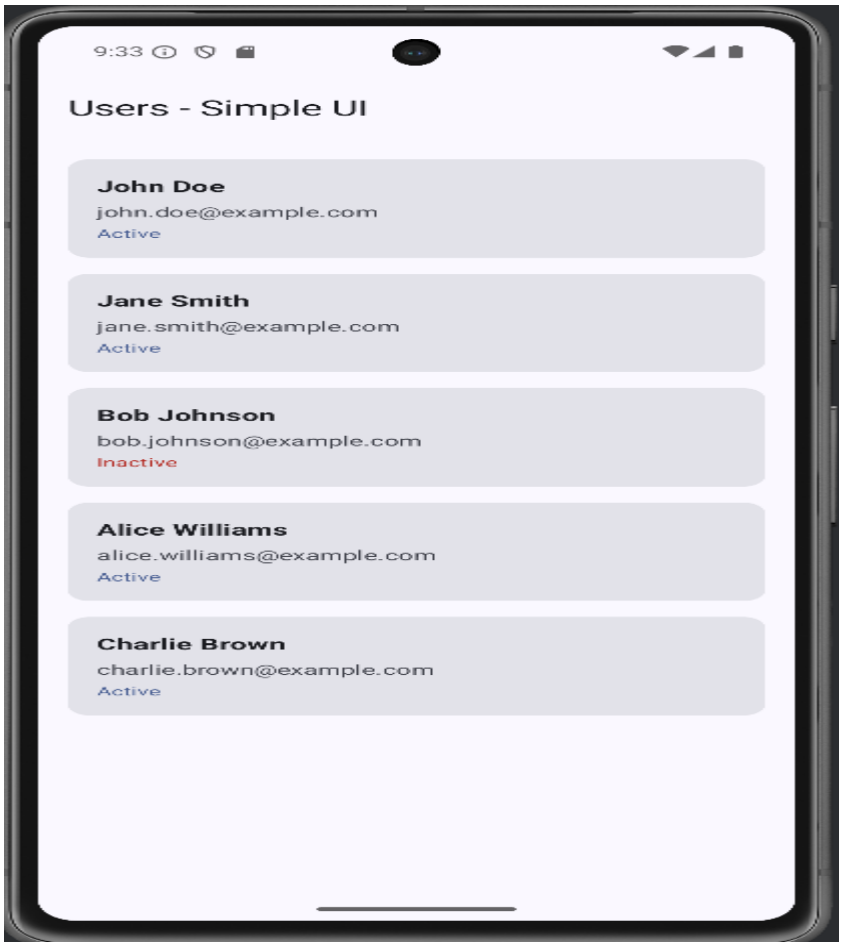
```
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.draw.clip
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import com.example.apparchitectureuilayer.data.model.User
import com.example.apparchitectureuilayer.presentation.screen.UserListScreenSimple

@Composable
fun AppNavigationSimple() {
    var currentUser by remember { mutableStateOf<User?>(null) }

    if (currentUser == null) {
        UserListScreenSimple(
            onUserClick = { userId ->
                // Find user and show details
                val users = listOf(
                    User(
                        id = "1",
                        name = "John Doe",
                        email = "john.doe@example.com",
                        avatar = null,
                        isActive = true
                    ),
                    User(
                        id = "2",
                        name = "Jane Smith",
                        email = "jane.smith@example.com",
                        avatar = null,
                        isActive = true
                    ),
                    User(
                        id = "3",
                        name = "Bob Johnson",
                        email = "bob.johnson@example.com",
                        avatar = null,
                        isActive = false
                    ),
                    User(
                        id = "4",
                        name = "Alice Williams",
                        email = "alice.williams@example.com",
                        avatar = null,
                        isActive = true
                    ),
                    User(
```

```
                id = "5",
                name = "Charlie Brown",
                email = "charlie.brown@example.com",
                avatar = null,
                isActive = true
            )
        )
        currentUser = users.find { it.id == userId }
    }
)
} else {
    UserDetailScreenSimple(
        user = currentUser!!,
        onBack = { currentUser = null }
    )
}
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun UserDetailScreenSimple(
    user: User,
    onBack: () -> Unit
) {
    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("User Details") },
                navigationIcon = {
                    Button(onClick = onBack) {
                        Text("← Back")
                    }
                }
            )
        }
    ) { padding ->
        Column(
            modifier = Modifier
                .fillMaxSize()
                .padding(padding)
                .padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(16.dp)
        ) {
            // User avatar placeholder
            Box(
                modifier = Modifier
                    .size(120.dp)
                    .align(Alignment.CenterHorizontally)
```

```
                .clip(RoundedCornerShape(16.dp))
                .background(
                    color = MaterialTheme.colorScheme.primaryContainer
                ),
            contentAlignment = Alignment.Center
        ) {
            Text(
                text = user.name.first().uppercase(),
                style = MaterialTheme.typography.headlineLarge,
                color = MaterialTheme.colorScheme.onPrimaryContainer
            )
        }

        // User info
        Card(
            modifier = Modifier.fillMaxWidth()
        ) {
            Column(
                modifier = Modifier
                    .fillMaxWidth()
                    .padding(16.dp),
                verticalArrangement = Arrangement.spacedBy(8.dp)
            ) {
                Text(
                    text = user.name,
                    style = MaterialTheme.typography.headlineMedium,
                    fontWeight = FontWeight.Bold
                )
                Text(
                    text = user.email,
                    style = MaterialTheme.typography.bodyLarge,
                    color = MaterialTheme.colorScheme.onSurfaceVariant
                )
                Text(
                    text = "Status: ${if (user.isActive) "Active" else "Inactive"}",
                    style = MaterialTheme.typography.bodyMedium,
                    color = if (user.isActive)
                        MaterialTheme.colorScheme.primary else
                        MaterialTheme.colorScheme.error
                )
                Text(
                    text = "ID: ${user.id}",
                    style = MaterialTheme.typography.bodySmall,
                    color = MaterialTheme.colorScheme.onSurfaceVariant
                )
            }
        }
    }
```

```
    }
}
```

Output


9:33
Users - Simple UI
John Doe
john.doe@example.com
Active
Jane Smith
jane.smith@example.com
Active
Bob Johnson
bob.johnson@example.com
Inactive
Alice Williams
alice.williams@example.com
Active
Charlie Brown
charlie.brown@example.com
Active

9:34
← Back
User Details
J
John Doe
john.doe@example.com
Status: Active
ID: 1

Expt No : 9 **Date:**

App Architecture (Persistence)

Aim

To store and retrieve data using Room persistence library.

Algorithm

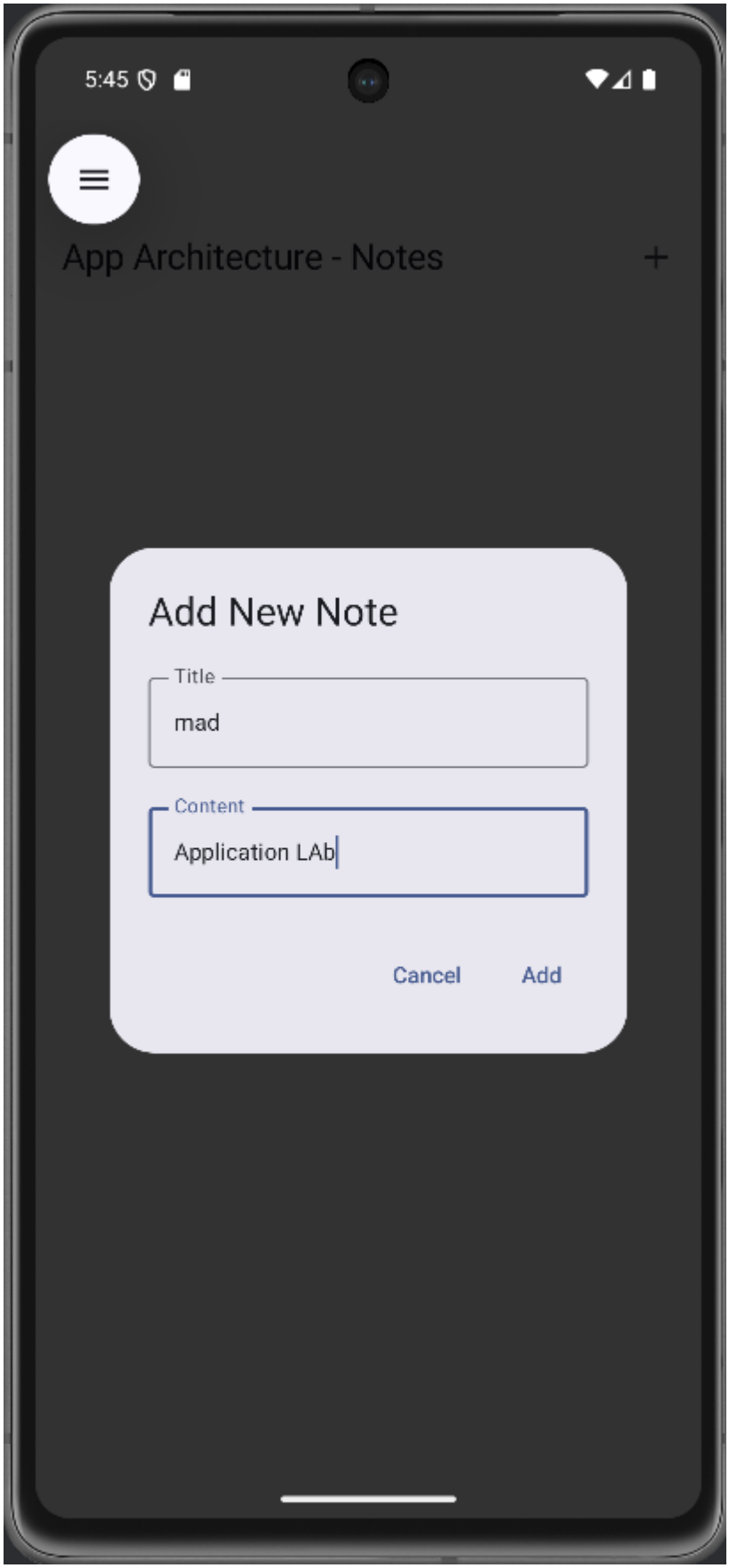
1. Add Room dependencies.
2. Create Entity class.
3. Create DAO interface.
4. Create Database class.
5. Perform CRUD operations.

MainActivity.kt

```
package com.example.apparchitecturepersistence
import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.foundation.layout.padding
import androidx.compose.material3.Scaffold
import androidx.compose.runtime.Composable
import androidx.compose.runtime.collectAsState
import androidx.compose.runtime.getValue
import androidx.compose.ui.Modifier
import androidx.compose.ui.tooling.preview.Preview
import androidx.lifecycle.viewmodel.compose.viewModel
import androidx.compose.runtime.livedata.observeAsState
import com.example.apparchitecturepersistence.data.NoteDatabase
import com.example.apparchitecturepersistence.data.Note
import com.example.apparchitecturepersistence.repository.NoteRepository
import
com.example.apparchitecturepersistence.ui.theme.AppArchitecturePersistenceTheme
import com.example.apparchitecturepersistence.ui.screens.NoteListScreen
import com.example.apparchitecturepersistence.viewmodel.NoteViewModel
import com.example.apparchitecturepersistence.viewmodel.NoteViewModelFactory

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()

        val database = NoteDatabase.getDatabase(this)
        val repository = NoteRepository(database.noteDao())
```

```
        setContent {
            AppArchitecturePersistenceTheme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    NoteApp(
                        repository = repository,
                        modifier = Modifier.padding(innerPadding)
                    )
                }
            }
        }
    }
}

@Composable
fun NoteApp(repository: NoteRepository, modifier: Modifier = Modifier) {
    val viewModel: NoteViewModel = viewModel(
        factory = NoteViewModelFactory(repository)
    )

    val notes by viewModel.allNotes.observeAsState(initial = emptyList())
    val selectedNote by viewModel.selectedNote.observeAsState()
    val isLoading by viewModel.isLoading.observeAsState(initial = false)
    val errorMessage by viewModel.errorMessage.observeAsState()

    NoteListScreen(
        notes = notes ?: emptyList(),
        selectedNote = selectedNote,
        isLoading = isLoading ?: false,
        errorMessage = errorMessage,
        onNoteClick = { note -> viewModel.selectNote(note) },
        onNoteDelete = { note -> viewModel.deleteNote(note) },
        onNoteToggle = { note -> viewModel.toggleNoteCompletion(note) },
        onAddNote = { title, content -> viewModel.insertNote(title, content) },
        onUpdateNote = { note -> viewModel.updateNote(note) },
        onClearError = { viewModel.clearError() },
        onDeleteAll = { viewModel.deleteAllNotes() },
        formatTimestamp = { timestamp -> viewModel.formatTimestamp(timestamp) },
        modifier = modifier
    )
}

@Preview(showBackground = true)
@Composable
fun NoteAppPreview() {
    AppArchitecturePersistenceTheme {
        // Preview would need mock repository for full functionality
    }
}
```

Output


5:45
App Architecture - Notes
Add New Note
Title
mad
Content
Application LAb
Cancel
Add

Expt No : 10 **Date:**

Advanced RecyclerView Use Cases

Aim
To display complex and dynamic lists using RecyclerView.

Algorithm
1. Add RecyclerView to layout.
2. Create ViewHolder.
3. Create Adapter.
4. Bind data list.
5. Handle item clicks.

```
package com.example.advancedrecyclerviewusecases

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.ItemTouchHelper
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView
import
com.example.advancedrecyclerviewusecases.adapter.AdvancedRecyclerViewAdapter
import
com.example.advancedrecyclerviewusecases.helper.ItemTouchHelperCallback
import com.example.advancedrecyclerviewusecases.model.DataProvider
import com.example.advancedrecyclerviewusecases.model.Item
import com.google.android.material.snackbar.Snackbar

class MainActivity : AppCompatActivity() {

    private lateinit var recyclerView: RecyclerView
    private lateinit var adapter: AdvancedRecyclerViewAdapter
    private lateinit var itemTouchHelper: ItemTouchHelper
    private var items = DataProvider.getSampleData()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        setupRecyclerView()
        setupSwipeToDelete()
        setupDragAndDrop()
    }
```

```
private fun setupRecyclerView() {
    recyclerView = findViewById(R.id.recyclerView)
    adapter = AdvancedRecyclerViewAdapter(
        items = items,
        onItemClick = { item ->
            handleItemClick(item)
        },
        onDeleteItem = { item ->
            handleDeleteItem(item)
        }
    )

    recyclerView.apply {
        layoutManager = LinearLayoutManager(this@MainActivity)
        adapter = this@MainActivity.adapter
    }
}

private fun handleItemClick(item: Item) {
    Snackbar.make(
        recyclerView,
        "Clicked: ${item.title}",
        Snackbar.LENGTH_SHORT
    ).show()
}

private fun handleDeleteItem(item: Item) {
    val position = items.indexOf(item)
    if (position != -1) {
        items = items.toMutableList().apply { removeAt(position) }
        adapter.removeItem(item)

        Snackbar.make(
            recyclerView,
            "Deleted: ${item.title}",
            Snackbar.LENGTH_LONG
        ).setAction("Undo") {
            items = items.toMutableList().apply { add(position, item) }
            adapter.updateItems(items)
        }.show()
    }
}

private fun setupSwipeToDelete() {
    val callback = ItemTouchHelperCallback(
        onItemMove = { fromPosition, toPosition ->
            // Handle item move for drag and drop
```

```
            val movedItem = items[fromPosition]
            items = items.toMutableList().apply {
                removeAt(fromPosition)
                add(toPosition, movedItem)
            }
            adapter.notifyItemMoved(fromPosition, toPosition)
        },
        onItemSwiped = { position ->
            // Handle item swipe for delete
            val swipedItem = items[position]
            handleDeleteItem(swipedItem)
        }
    )

    itemTouchHelper = ItemTouchHelper(callback)
    itemTouchHelper.attachToRecyclerView(recyclerView)
  }

  private fun setupDragAndDrop() {
    // Drag and drop is already set up in setupSwipeToDelete()
    // The ItemTouchHelper handles both swipe and drag operations
  }
}
```

Output

Expt No :1 1 **Date:**

Connect to the Internet

Aim

To fetch data from internet using REST API.

Algorithm

1. Add internet permission.
2. Add Retrofit dependency.
3. Create API interface.
4. Make network call.
5. Display response data.

Coding

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">
<application
android:allowBackup="true"
android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:supportsRtl="true"
android:theme="@style/Theme.SendEmail"
tools:targetApi="31">
<activity
android:name=".MainActivity"
android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
```

```
tools:context=".MainActivity">
<TextView
android:id="@+id/tvEmail"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="To" />
<EditText
android:id="@+id/etEmail"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvSubject"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Subject" />
<EditText
android:id="@+id/etSubject"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvMessage"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Message" />
<EditText
android:id="@+id/etMessage"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName" />
<Button
android:id="@+id/btSend"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Send"
android:textAllCaps="false" />
</LinearLayout>
```

MainActivity.kt

```
package org.rajalakshmi.sendemail
import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
```

```
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
val etEmail : EditText = findViewById(R.id.etEmail)
val etSubject : EditText = findViewById(R.id.etSubject)
val etMessage : EditText = findViewById(R.id.etMessage)
val btSend : Button = findViewById(R.id.btSend)
btSend.setOnClickListener {
val email = etEmail.text.toString()
val subject = etSubject.text.toString()
val message = etMessage.text.toString()
val intent = Intent(Intent.ACTION_SEND)
intent.putExtra(Intent.EXTRA_EMAIL, arrayOf(email))
intent.putExtra(Intent.EXTRA_SUBJECT, subject)
intent.putExtra(Intent.EXTRA_TEXT, message)
intent.type = "message/rfc822"
startActivity(Intent.createChooser(intent, "Choose an Email client
:"))
}
}
}
```

Output

Expt No : 12 **Date:**

Repository Pattern and WorkManager

Aim

To manage data sources and background tasks efficiently.

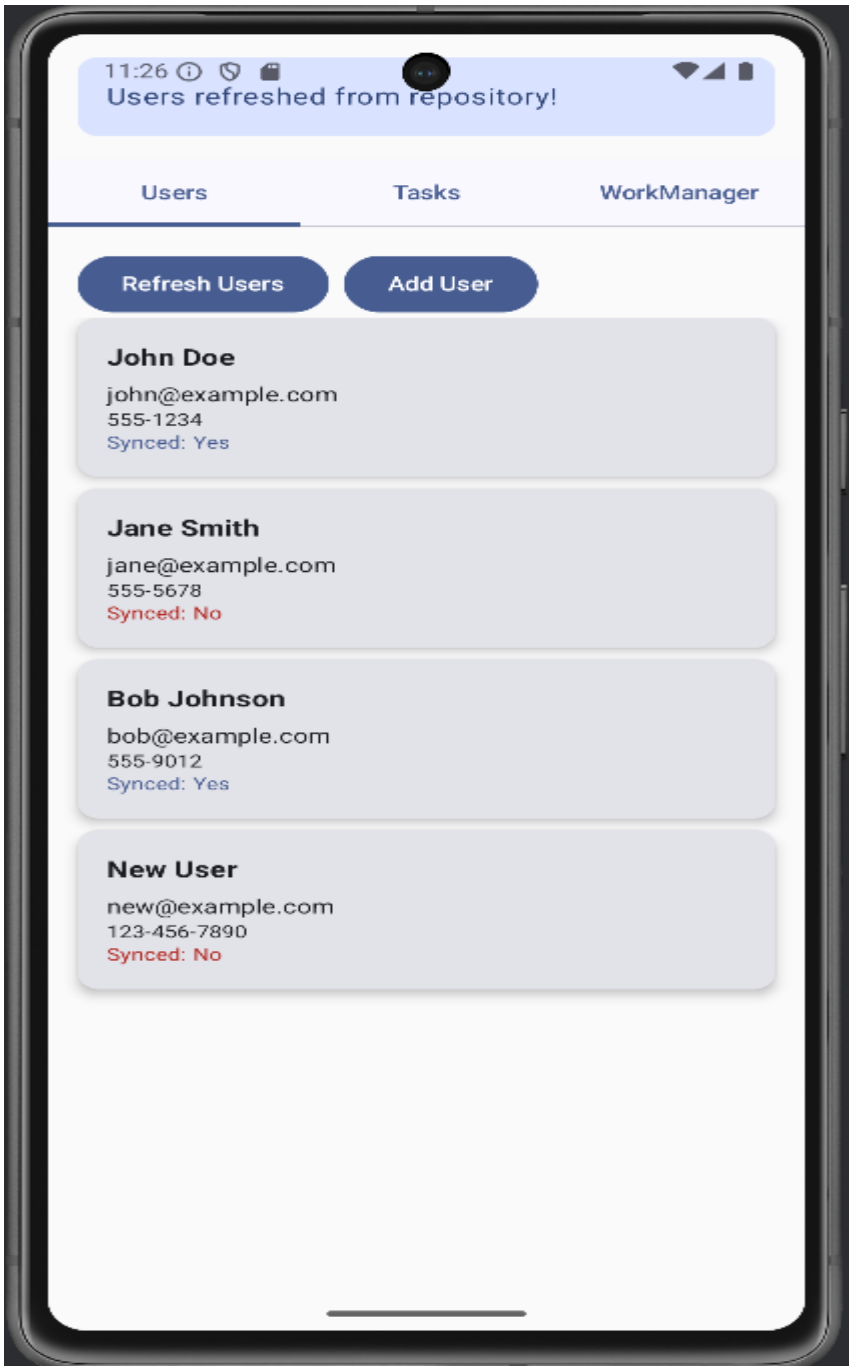
Algorithm

1. Create repository class.
2. Connect repository to ViewModel.
3. Add WorkManager dependency.
4. Create Worker class.
5. Schedule background task.

Coding

```
package com.example.repository

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import com.example.repository.ui.theme.RepositoryTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            RepositoryTheme {
                RepositoryDemoScreen()
            }
        }
    }
}
```

Output


11:26
Users refreshed from repository!
Users
Tasks
WorkManager
Refresh Users
Add User
John Doe
john@example.com
555-1234
Synced: Yes
Jane Smith
jane@example.com
555-5678
Synced: No
Bob Johnson
bob@example.com
555-9012
Synced: Yes
New User
new@example.com
123-456-7890
Synced: No

Expt No : 13 **Date:**

App UI Design

Aim
To design attractive, responsive, and user-friendly Android UI.

Algorithm
1. Apply Material Design principles.
2. Use themes and styles.
3. Add colors and fonts.
4. Use icons and animations.
5. Test UI responsiveness.

MainActivity.kt

```
package com.example.japp9

import android.os.Bundle
import android.os.Handler
import android.os.Looper
import android.text.Editable
import android.text.TextWatcher
import android.widget.Button
import android.widget.EditText
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView
import com.example.japp9.adapter.AdvancedRecyclerViewAdapter
import com.example.japp9.model.ExpandableItemModel
import com.example.japp9.model.RecyclerViewItem
import com.example.japp9.model.User
import com.example.japp9.utils.DragAndDropCallback
import com.example.japp9.utils.FilterUtils
import com.example.japp9.utils.InfiniteScrollListener
import com.example.japp9.utils.SwipeToDeleteCallback

class MainActivity : AppCompatActivity() {

    private lateinit var recyclerView: RecyclerView
    private lateinit var adapter: AdvancedRecyclerViewAdapter
    private lateinit var searchEditText: EditText
    private lateinit var toggleSwipeButton: Button
```

```
    private lateinit var toggleDragButton: Button

    private var allItems = mutableListOf<RecyclerViewItem>()
    private var currentItems = mutableListOf<RecyclerViewItem>()
    private var swipeToDeleteHelper: androidx.recyclerview.widget.ItemTouchHelper? =
null
    private var dragAndDropHelper: androidx.recyclerview.widget.ItemTouchHelper? =
null
    private var infiniteScrollListener: InfiniteScrollListener? = null

    private var isLoading = false
    private var currentPage = 1
    private val itemsPerPage = 10

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        initViews()
        setupRecyclerView()
        generateInitialData()
        setupListeners()
        setupGestures()
    }

    private fun initViews() {
        recyclerView = findViewById(R.id.recyclerView)
        searchEditText = findViewById(R.id.searchEditText)
        toggleSwipeButton = findViewById(R.id.toggleSwipeButton)
        toggleDragButton = findViewById(R.id.toggleDragButton)
    }

    private fun setupRecyclerView() {
        adapter = AdvancedRecyclerViewAdapter(
            items = currentItems,
            onItemClick = { item ->
                handleItemClick(item)
            },
            onDeleteItem = { position ->
                adapter.removeItem(position)
            }
        )

        recyclerView.layoutManager = LinearLayoutManager(this)
        recyclerView.adapter = adapter

        // Setup infinite scrolling
        val layoutManager = recyclerView.layoutManager as LinearLayoutManager
```

```
        infiniteScrollListener = object : InfiniteScrollListener(layoutManager) {
            override fun onLoadMore() {
                loadMoreItems()
            }
        }
        recyclerView.addOnScrollListener(infiniteScrollListener!!)
    }

    private fun generateInitialData() {
        // Generate sample users
        val users = (1..20).map { index ->
            User(
                id = index,
                name = "User $index",
                email = "user$index@example.com",
                avatar = "",
                isActive = index % 2 == 0
            )
        }

        // Generate sample expandable items
        val expandableItems = (1..5).map { index ->
            ExpandableItemModel(
                id = index + 100,
                title = "Category $index",
                description = "Description for category $index",
                subItems = (1..3).map { subIndex -> "Sub item $subIndex in category $index" }
            )
        }

        allItems.clear()
        allItems.add(RecyclerViewItem.HeaderItem("Active Users"))
        users.filter { it.isActive }.take(5).forEach {
            allItems.add(RecyclerViewItem.UserItem(it))
        }

        allItems.add(RecyclerViewItem.HeaderItem("Categories"))
        expandableItems.forEach {
            allItems.add(RecyclerViewItem.ExpandableItem(it))
        }

        allItems.add(RecyclerViewItem.HeaderItem("All Users"))
        users.forEach {
            allItems.add(RecyclerViewItem.UserItem(it))
        }

        currentItems.clear()
        currentItems.addAll(allItems.take(itemsPerPage))
```

```
        adapter.updateItems(currentItems)
    }

    private fun setupListeners() {
        searchEditText.addTextChangedListener(object : TextWatcher {
            override fun beforeTextChanged(s: CharSequence?, start: Int, count: Int, after: Int)
{}
            override fun onTextChanged(s: CharSequence?, start: Int, before: Int, count: Int)
{}
            override fun afterTextChanged(s: Editable?) {
                filterItems(s?.toString() ?: "")
            }
        })

        toggleSwipeButton.setOnClickListener {
            toggleSwipeToDelete()
        }

        toggleDragButton.setOnClickListener {
            toggleDragAndDrop()
        }
    }

    private fun setupGestures() {
        // Setup swipe to delete
        val swipeCallback = SwipeToDeleteCallback { position ->
            adapter.removeItem(position)
        }
        swipeToDeleteHelper =
androidx.recyclerview.widget.ItemTouchHelper(swipeCallback)
        swipeToDeleteHelper?.attachToRecyclerView(recyclerView)

        // Setup drag and drop
        val dragCallback = DragAndDropCallback { fromPosition, toPosition ->
            adapter.moveItem(fromPosition, toPosition)
        }
        dragAndDropHelper =
androidx.recyclerview.widget.ItemTouchHelper(dragCallback)
        // Don't attach by default, let user toggle it
    }

    private fun handleItemClick(item: RecyclerViewItem) {
        when (item) {
            is RecyclerViewItem.ExpandableItem -> {
                // Toggle expandable item
                val updatedItem = item.item.copy(isExpanded = !item.item.isExpanded)
                val index = currentItems.indexOf(item)
                if (index != -1) {
```

```
                currentItems[index] = RecyclerViewItem.ExpandableItem(updatedItem)
                adapter.notifyItemChanged(index)
            }
        }
        is RecyclerViewItem.UserItem -> {
            // Handle user click - could show details or edit
            println("Clicked on user: ${item.user.name}")
        }
        else -> {}
    }
}

private fun filterItems(query: String) {
    val filtered = FilterUtils.filterItems(allItems, query)
    currentItems.clear()
    currentItems.addAll(filtered.take(itemsPerPage))
    adapter.updateItems(currentItems)
}

private fun toggleSwipeToDelete() {
    if (swipeToDeleteHelper == null) {
        val swipeCallback = SwipeToDeleteCallback { position ->
            adapter.removeItem(position)
        }
        swipeToDeleteHelper =
androidx.recyclerview.widget.ItemTouchHelper(swipeCallback)
        swipeToDeleteHelper?.attachToRecyclerView(recyclerView)
        toggleSwipeButton.text = "Disable Swipe"
    } else {
        swipeToDeleteHelper?.attachToRecyclerView(null)
        swipeToDeleteHelper = null
        toggleSwipeButton.text = "Enable Swipe"
    }
}

private fun toggleDragAndDrop() {
    if (dragAndDropHelper == null) {
        val dragCallback = DragAndDropCallback { fromPosition, toPosition ->
            adapter.moveItem(fromPosition, toPosition)
        }
        dragAndDropHelper =
androidx.recyclerview.widget.ItemTouchHelper(dragCallback)
        dragAndDropHelper?.attachToRecyclerView(recyclerView)
        toggleDragButton.text = "Disable Drag"
    } else {
        dragAndDropHelper?.attachToRecyclerView(null)
        dragAndDropHelper = null
        toggleDragButton.text = "Enable Drag"
```

```
        }
    }

    private fun loadMoreItems() {
        if (isLoading) return

        isLoading = true

        // Add loading indicator
        val loadingList = currentItems.toMutableList()
        loadingList.add(RecyclerViewItem.LoadingItem())
        adapter.updateItems(loadingList)

        // Simulate network delay
        Handler(Looper.getMainLooper()).postDelayed({
            val startIndex = currentPage * itemsPerPage
            val endIndex = (startIndex + itemsPerPage).coerceAtMost(allItems.size)

            if (startIndex < allItems.size) {
                currentItems.addAll(allItems.subList(startIndex, endIndex))
                currentPage++

                // Remove loading indicator and add new items
                adapter.updateItems(currentItems)
                infiniteScrollListener?.setLoaded()
            }

            isLoading = false
        }, 1500)
    }
}
```

10:05
App UI Designer
Components
Button
Text Field
Text
Image
Card
Contain er
Elements
Properties
Select an element to edit properties

10:07
App UI Designer
Components
Button
Text Field
Text
Image
Card
Contain er
Elements
BUTTON (element_1)
Properties
Text
Button
Widt h
100
Heig ht
50
Corner Radius
8
R e d
B l u e
G r e e n